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Source-Stratified Evaluation of Out-of-Distribution Detection in Fruit-Quality Classification Under Species and Treatment-Labelled Shifts

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Abstract

Automated fruit-quality classification using computer vision can achieve strong performance on familiar image distributions but may become unreliable under changes in data sources, fruit species, and post-harvest treatment conditions. This study presents a source-stratified empirical evaluation of a frozen MobileNetV2-based six-class fruit-quality classifier and multiple out-of-distribution (OOD) detection approaches. The evaluation uses 13,465 image records from three image repositories, organized into primary in-distribution, external known-class, unseen-species, treatment-labelled, and combined-shift cohorts.

Five post-hoc OOD scoring methods—Maximum Softmax Probability (MSP), Energy, an ODIN-style perturbation variant, Ledoit–Wolf Mahalanobis distance, and a modified OpenMax-style heuristic—are compared alongside an operational Energy–MSP dual-gate cascade. Closed-set accuracy decreases from 96.92% on the primary in-distribution test partition to 68.36% on the pooled external known-class partition, reflecting a 28.56 percentage-point difference across evaluation sources. Within the FruitVision repository, Energy achieves an AUROC of 81.57% for the known-versus-unseen-species comparison, with a standalone Energy operational false acceptance rate of 13.18% under its locked validation threshold, whereas the observed AUROC on the source-matched Sultana unseen-species comparison was 67.52%. In the evaluated FruitVision treatment-labelled cohort, discrimination between natural known-class images and treatment-labelled known-class images is limited, with Energy achieving 53.13% AUROC. Under fixed dual-gate operating thresholds calibrated on validation data, 42.83% of the treatment-labelled image records are accepted by the evaluated gate and assigned one of the classifier's Fresh-class output labels.

The findings demonstrate that post-hoc OOD detection performance varies substantially across distribution shifts and data sources. In the evaluated treatment cohort, tested RGB post-hoc scoring methods did not reliably separate images recorded under the dataset-reported treatment protocol from natural fruit. The study provides an image-level benchmark analysis with source-stratified comparisons, uncertainty estimates, duplicate-sensitivity analysis, and explicit limitations concerning dataset labels, specimen independence, and operational validation.

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Keywords: Fruit quality classification; Open-set recognition; Out-of-distribution detection; Domain shift; MobileNetV2; Energy-based detection; Image classification; Food quality assessment

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1. Introduction

1.1. Research Background and Motivation

Post-harvest agricultural loss represents a critical challenge for global food security, economic stability, and supply chain sustainability [1]. According to the Food and Agriculture Organization (FAO), approximately 14% of global food production is lost between harvest and retail distribution [1]. In regional agrarian economies, post-harvest deterioration across domestic transit corridors is particularly acute. For example, nationwide surveys in India conducted by the Indian Council of Agricultural Research (ICAR-CIPHET) indicate that post-harvest losses of commercial fruits consistently range between 6.70% and 15.88% (e.g., 10.39% in apple, 7.76% in banana, and 15.88% in citrus) [2]. These losses disproportionately penalize smallholder farmers and aggregation packhouses, where produce sorting and quality grading have historically relied upon manual, visual inspection. Manual grading is inherently subjective, labour-intensive, prone to operator fatigue, and incapable of scaling to high-throughput sorting facilities [3, 4].

In response to these operational bottlenecks, computer vision and deep learning have emerged as promising non-destructive methodologies for automated fruit freshness assessment, ripeness grading, and surface defect classification [3–6]. Convolutional neural networks (CNNs), particularly lightweight architectures such as MobileNetV2 [7], have become central to practical agricultural deployments. Depthwise separable convolutions and inverted residual bottlenecks allow MobileNetV2 to execute on resource-constrained mobile phones and edge sorting devices with parameter budgets under 4 million weights [5, 7]. In controlled academic evaluations on curated benchmarks, reported top-1 accuracies often exceed 95% on specific multi-class produce datasets [4–6].

1.2. The Closed-Set Assumption and Its Vulnerabilities in Agricultural Vision

Despite high benchmark accuracy, practical deployment of deep learning models in real-world agricultural environments encounters fundamental reliability barriers. Standard supervised image classifiers operate under a strict closed-set assumption: the training, validation, and operational deployment data are presumed to be independent and identically distributed (i.i.d.) over a fixed, finite set of semantic classes, $\mathcal{Y} = \{1, 2, \ldots, K\}$ [8, 9]. In practical agricultural workflows, however, this assumption rarely holds. Vision models deployed at packhouses, retail terminals, or farm collection centres confront three distinct, compounding categories of distribution shift:

1. Covariate and Domain Shift: Images collected across external facilities exhibit variations in background illumination, sensor optics, specular glare, ambient shadows, and post-harvest handling marks that differ from training sets [3, 5, 10, 11]. Furthermore, natural biological diversity in produce appearance, surface texture, and pigmentation across agricultural batches can induce domain shifts that affect learned feature representations [11, 12].
2. Semantic Shift (Unseen Species): In automated sorting lines, bins, or field crates, models trained exclusively on a small palette of target crops (e.g., apple, banana, and orange) frequently encounter non-target produce species (e.g., mango, guava, jujube, pomegranate, or lime). Under standard closed-set inference, deep networks lack an explicit rejection mechanism; they are forced to assign an unseen species to one of the K known classes, often with uncalibrated, erroneously high confidence [8, 9].
3. Treatment-Labelled Condition Shift: In certain agricultural supply-chain and research contexts, produce may be subjected to chemical preservative treatments or immersion protocols intended to retard visible senescence. In curated benchmarks such as FruitVision [12], images are documented under a specific dataset-reported post-harvest treatment protocol involving immersion in a 10% formalin and 90% water

solution for 30 minutes, followed by drying and photography 48 hours later. Whether visual models trained on natural produce can reliably distinguish images recorded under such post-harvest treatment protocols remains an empirical question.

Under standard softmax classification, neural networks are notoriously overconfident [8]. Logit outputs normalized via the softmax function produce probability distributions that often peak sharply near 1.0 even when the input image is heavily corrupted, semantically anomalous, or completely out-of-distribution [8, 9]. In automated quality sorting, an overconfident false-positive classification—such as misclassifying an unseen non-target species or an anomalous produce lot as a fresh known-fruit class—presents substantial commercial and quality-assurance liabilities.

1.3. Open-Set Recognition and Out-of-Distribution Detection Foundations

To address closed-set failure modes, computer vision research has developed two closely related paradigms: Open Set Recognition (OSR) [13, 14] and Out-of-Distribution (OOD) Detection [9, 15].

The concept of open-set recognition was formalized by Scheirer et al. [13], who defined the objective as simultaneously classifying known training classes while bounding the open space risk—the risk of incorrectly labeling an unknown, novel class sample as any known class. Bendale and Boult [14] extended this concept to deep neural networks through OpenMax, which estimates the probability of an input belonging to an unknown class by fitting Weibull distributions to activation vectors in the penultimate layer.

Simultaneously, the broader machine learning community formalized generalized out-of-distribution detection [9, 15]. Hendrycks and Gimpel [9] established the Maximum Softmax Probability (MSP) baseline, showing that correctly classified in-distribution samples typically exhibit higher maximum posterior probabilities than anomalous inputs. Liang et al. [16] developed ODIN, which amplifies the separation between in-distribution and out-of-distribution samples using temperature-scaled softmax outputs and input-space gradient perturbations. Lee et al. [17] introduced a feature-space Mahalanobis distance metric that models class-conditional Gaussian distributions across intermediate hidden representations. More recently, Liu et al. [18] proposed Energy-based OOD detection, utilizing the log-sum-exp of the classifier logits to circumvent the probability saturation that characterizes standard softmax normalization [15, 18]. Additionally, Vaze et al. [19] demonstrated that closed-set classifier capacity and representation quality strongly dictate open-set discrimination performance.

In operational deployment, OOD detectors are typically implemented either as continuous ranking functions or as discrete rejection gates governed by a pre-calibrated threshold τ. When an input sample produces an anomaly score that violates the threshold, the system abstains from classification, triggering manual inspection or secondary non-optical assay.

1.4. Prior Agricultural OSR/OOD Studies and Specific Research Gaps

Open-set recognition and anomaly detection have recently attracted growing attention in agricultural vision. Several recent investigations have examined related challenges:

- Sattar et al. [20] developed SensorNet, combining computer vision, deep learning, and chemical sensors to identify chemical adulteration in fruit specimens.
- Bijoy et al. [12] introduced FruitVision, an image dataset containing fresh, rotten, and formalin-mixed fruit images.
- Wang et al. [21] proposed lightweight attention networks with unsupervised domain adaptation for cross-domain fruit classification.
- Pehlivan and Kara [22] introduced a lightweight one-shot open-set metric learning framework for food recognition in smart appliances.
- Jansi and Kavitha [23] developed PEOS-Net, a prototype-energy open-set network for robust crop leaf disease recognition beyond the closed-world assumption.

- Yan et al. [24] formulated PAG-MSBNet, using prototype-aware gated multi-scale branch networks for open-set crop disease recognition.

While these studies highlight the importance of non-standard conditions in agricultural classification, existing research has not systematically evaluated how established post-hoc OOD scoring functions behave when a frozen classifier faces multi-source external distributions. Specifically, four key empirical gaps remain:

1. Gap 1: Limited Evaluation Beyond Dataset-Specific Closed-Set Splits: Existing fruit-freshness studies predominantly report supervised classification performance on random train/test partitions within a single, curated dataset [4–6]. Such closed-set protocols evaluate neither model behavior on external image repositories nor discrimination against out-of-distribution instances. Furthermore, recent evaluations of agricultural image collections indicate that random partitioning within single repositories can allow near-duplicate images across splits, yielding optimistic closed-set performance estimates that do not reflect external test conditions [10, 11].
2. Gap 2: Cross-Repository Confounding in OOD Benchmarks: In open-set evaluations where out-of-distribution cohorts are drawn from a different repository than the known-class training data, differences in camera hardware, ambient illumination, and background context present a substantial methodological risk. Under such unstratified designs, repository-level acquisition artifacts can be confounded with true semantic divergence between fruit species [15]. Evaluating known and unknown categories collected under shared repository conditions (source matching) reduces repository-level confounding, though potential within-repository capture variations may remain.
3. Gap 3: Empirical Behavior Under Treatment-Labelled Shifts: While multimodal sensing has explored chemical adulteration detection combining optical imaging with specialized gas sensors [20], the behavior of post-hoc OOD scoring methods on a standard frozen RGB classifier facing dataset-reported treatment conditions [12] without auxiliary chemical sensors warrants systematic empirical investigation.
4. Gap 4: Distinguishing Retrospective Ranking from Operational Threshold Calibration: Many OOD detection studies primarily report retrospective metrics: AUROC assesses ranking discrimination across all decision thresholds without selecting an operating point, whereas FPR@95 evaluates the false positive rate at a retrospective threshold chosen post-hoc to achieve 95% true positive rate on the test cohort [9, 15, 18]. While suitable for relative algorithm comparison, retrospective metrics do not reflect operational deployment, where a discrete acceptance threshold must be pre-calibrated on validation data and held fixed during external evaluation. Measuring operational Unknown False Acceptance Rates (UFAR) alongside Known-Class Rejection Rates (KRR) under fixed validation thresholds provides complementary, deployment-context measurements of operational trade-offs.

1.5. Research Objectives and Formal Research Questions

This investigation addresses these gaps by conducting an extensive, source-stratified empirical evaluation of out-of-distribution detection in fruit image classification. The study evaluates a frozen MobileNetV2 six-class visual classifier across $N = 13,465$ image records curated from three distinct repositories under the Stage 4.0.3 external evaluation protocol.

The paper specifically investigates three core research questions:

- Research Question 1 (RQ1 — External Generalization): How does the closed-set classification performance (top-1 accuracy, macro-F1, and balanced accuracy) of a lightweight MobileNetV2 model change when evaluated on external repositories containing the same known fruit classes?
- Research Question 2 (RQ2 — OOD Discrimination): How effectively do post-hoc OOD scoring mechanisms (MSP, Energy, an ODIN-style perturbation variant, Ledoit–Wolf Mahalanobis distance, and a modified OpenMax-style heuristic) discriminate between in-distribution produce and distribution shifts when evaluated in source-matched cohorts examining (a) unseen-species semantic shift, (b) treatment-labelled condition shift, and (c) combined species-and-treatment shifts?

- Research Question 3 (RQ3 — Operational Reliability): How do fixed validation-calibrated rejection thresholds (τ_{95}) affect out-of-distribution false acceptance rates (UFAR) and known-class rejection rates (KRR) under external domain and condition shifts? (Where known-class retention is defined as $1 - \text{KRR}$.)

1.6. Evidence-Supported Research Contributions

The primary contribution of this work is an empirical source-stratified evaluation, rigorous uncertainty audit, and diagnostic failure analysis. The study does not propose a novel standalone OOD algorithm, nor does it claim real-world agricultural supply chain deployment. The four specific, evidence-backed contributions of this study are:

1. Audited Source-Stratified Benchmark Census ($N = 13,465$): We establish an audited image-level evaluation benchmark comprising 13,465 image records structured across five distinct partitions: Partition A (Primary ID Test, $N = 227$), Partition B (External Known-Class Shift, $N = 5,350$), Partition C (Treatment-Labelled Condition Shift, $N = 1,917$), Partition D1 (Unseen-Species Shift, $N = 4,776$), and Partition D2 (Combined Shift, $N = 1,195$). We incorporate exact-duplicate handling and perceptual-hash (pHash) sensitivity analysis across Hamming distance radii $r \in \{0, 4, 8\}$.
2. Quantification of Closed-Set Performance Differences Across Partitions: We evaluate the frozen MobileNetV2 classifier across partitions, observing a 28.56 percentage-point difference in top-1 accuracy between the primary in-distribution test partition (Partition A: 96.92% [94.71%, 98.68%]) and the pooled external known-class partition (Partition B: 68.36% [67.20%, 69.40%]), with balanced accuracy decreasing from 97.09% to 68.66%. Through source stratification, we show that performance varies across external repositories (FruitVision B: 69.14% accuracy, 69.56% balanced accuracy; Sultana B: 65.63% accuracy, 65.55% balanced accuracy), with closed-set errors heavily concentrated in the fresh apple class (where top-1 accuracy drops to 32.12% across pooled external images due to confusion with rotten apple and rotten orange).
3. Source-Matched Evaluation Across Semantic and Treatment Cohorts: By benchmarking within shared repositories, source matching reduces cross-repository confounding while potential within-repository capture variation remains. For the Within-FruitVision known-versus-unseen-species comparison (Cohort A), Energy-based detection achieves an AUROC of 81.57% [80.81%, 82.38%] with an operational UFAR of 13.18% [12.07%, 14.34%] ($366/2,776$ image records at $\tau_{\text{Energy}} = 1.054579$), whereas discrimination was lower on the Within-Sultana comparison (Cohort D: Energy AUROC of 67.52% [65.67%, 69.25%], operational UFAR of 34.15% [32.35%, 35.90%], $683/2,000$ image records).
4. Empirical Evaluation of Treatment-Labelled Images and Operational Trade-Offs: In the evaluated FruitVision treatment-labelled cohort (Cohort B), the frozen classifier and tested post-hoc OOD scoring methods did not reliably distinguish images labelled under the dataset-reported treatment condition from the natural known-class cohort, with Energy achieving an AUROC of 53.13% [52.16%, 54.09%]. Under locked validation operating thresholds (τ_{95}), the Dual-Gate cascade yields an operational UFAR-Fresh of 42.83% [41.26%, 44.34%] ($821/1,917$ image records), accepting and assigning one of the classifier's Fresh-class output labels to these image records. Furthermore, for the Dual-Gate cascade under τ_{95}, an operational UFAR of 16.07% [14.77%, 17.40%] ($446/2,776$ image records) on FruitVision unseen species incurs an external known-class rejection rate (KRR) of 41.98% [40.49%, 43.49%] ($1,745/4,157$ image records, corresponding to a retention rate of $1 - \text{KRR} \approx 58.02\%$) on FruitVision Known, and 42.43% [41.10%, 43.76%] ($2,270/5,350$ image records) across pooled external known images (Partition B).

2. Related Work

This section reviews the scientific literature foundational to the present study across four key domains: (1) computer vision methodologies for fruit quality assessment and freshness classification; (2) cross-domain

generalization and domain adaptation in agricultural vision systems; (3) established open-set recognition and out-of-distribution (OOD) detection algorithms; and (4) recent applications of open-set and anomaly detection frameworks to food and agricultural imagery. The section concludes with a structured comparison matrix and an explicit definition of the research gap addressed by this study.

2.1. Computer Vision for Fruit Quality Classification

Automated post-harvest quality assessment has progressed significantly over the past two decades, transitioning from classical hand-crafted optical feature extraction to deep convolutional neural networks (CNNs) [3]. Early automated sorting systems relied predominantly on engineered representations of colour, morphological geometry, and surface texture [3]. Colour histograms computed across RGB, HSV, and CIE $L^*a^*b^*$ spaces served as primary indicators of commercial maturity, chlorophyll degradation, and surface pigmentation changes. Morphological descriptors—including planar area, eccentricity, circularity, and perimeter irregularity—were leveraged to identify mechanical malformations and physical grading deviations. Texture descriptors, such as Gray-Level Co-occurrence Matrices (GLCM) and local binary patterns (LBP), provided statistical representations of surface coarseness, fungal lesions, and skin shrivelling. However, these classical pipelines exhibited acute vulnerability to illumination variations, background reflections, camera sensor discrepancies, and natural biological phenotypic diversity within identical cultivars [3].

The advent of deep convolutional architectures enabled end-to-end hierarchical representation learning, capturing subtle non-linear manifestations of produce senescence, mechanical bruising, and microbial decay without manual feature engineering [3, 4]. Multi-task convolutional networks have demonstrated the ability to concurrently estimate fruit categories, ripeness stages, and defect severity levels from unified photographic inputs [4]. Specialized deep architectures have achieved high classification accuracy on curated benchmarks for distinguishing fresh from rotten produce across common commercial species [5, 6].

In post-harvest logistics, packing facilities, and farm-gate sorting stations, computational efficiency and hardware constraints represent primary deployment criteria. High-throughput sorting conveyors and handheld mobile grading devices cannot accommodate computationally intensive, high-parameter vision backbones. Consequently, lightweight architectures have become the standard for embedded agricultural vision [5]. In particular, the MobileNetV2 architecture [7] has been widely adopted due to its use of depthwise separable convolutions, inverted residual blocks, and linear bottleneck layers. With approximately 2.26×10^6 trainable parameters (at standard width multiplier $\alpha = 1.0$), MobileNetV2 dramatically curtails floating-point operations while preserving high discriminative capacity. Deep transfer learning—wherein backbones pre-trained on large-scale natural image datasets (such as ImageNet) are fine-tuned on target agricultural domains—has further enabled robust feature extraction even when localized training sets are constrained in sample size [5, 6].

Despite high reported accuracies under closed-set evaluation protocols, photographic image acquisition in practical agricultural environments introduces severe distributional challenges. Ambient lighting fluctuations, direct sunlight glint, variable artificial warehouse illumination, diverse photographic backgrounds, and camera sensor colour-balance differences introduce pervasive domain shift [10, 11]. The FruitQ repository introduced by Abayomi-Alli et al. [10] provided a structured multi-fruit dataset comprising 11 fruit varieties categorized into three freshness states (Fresh, Mild, and Rotten), evaluating five baseline deep learning architectures (ShuffleNet, SqueezeNet, EfficientNet, ResNet18, and MobileNetV2) under closed-set protocols.

The empirical evaluation conducted in the present study examines photographic produce records across multiple repositories, drawing upon two primary external dataset publications:

1. Sultana et al. (2022) [11]: Introduced a photographic dataset for recognizing fresh and rotten fruit specimens (Data in Brief, 44:108552). The repository encompasses 16 fresh and rotten categories representing eight fruit types: grape, guava, jujube, pomegranate, strawberry, apple, banana, and orange. Images were collected from retail fruit shops and authentic agricultural field environments under natural ambient lighting, with collection and condition labelling conducted with assistance from an agricultural domain specialist.

2. Bijoy et al. (2025) [12]: Introduced FruitVision, a benchmark dataset for fresh, rotten, and formalin-mixed fruit detection (Data in Brief, 61:111752). The dataset comprises five fruit species (apple, banana, mango, orange, and grape) evaluated across three dataset-reported visual conditions: fresh, rotten, and a treatment-labelled condition (immersion in a 10% formalin and 90% water solution for 30 minutes, followed by 48-hour drying and photography, as documented by the dataset authors). The present study evaluates these photographic records under their dataset-reported labels and does not conduct chemical assays or independently verify formalin residue. Image collection was conducted using three distinct mobile smartphone cameras across varied indoor and outdoor lighting environments. The published raw collection comprises 10,154 original photographs, with geometric and photometric augmentations expanding the published archive to 81,232 images.

Evaluation Census Clarification: We emphasize that the published raw and augmented archive sizes reported by dataset authors (e.g., the 10,154 raw or 81,232 augmented images in FruitVision) represent upstream repository totals. These figures must not be confused with the audited benchmark cohort evaluated in the present study. As detailed in Section 3, our evaluation utilizes an audited local census of exactly 13,465 evaluated image records (Primary Kaggle: 227; FruitVision: 10,045; Sultana: 3,193) across five dedicated partitions. Every evaluated image record is uniquely tracked through cryptographic hashes, exact duplicates are systematically audited, and perceptual near-duplicates are evaluated via sensitivity analysis across Hamming distance thresholds. We explicitly note that image records cannot be assumed to represent independent biological specimens.

2.2. Cross-Domain and External Generalization in Fruit Classification

Supervised machine learning algorithms are traditionally grounded in the closed-set, independent and identically distributed (i.i.d.) assumption, wherein training and test samples are drawn from an identical underlying probability distribution: $P_{\text{train}}(X, Y) = P_{\text{test}}(X, Y)$ [15]. In practical agricultural deployments, however, this assumption rarely holds due to diverse distributional shifts. To analyze these phenomena rigorously, we maintain precise distinctions among shift mechanisms [15]:

- Covariate Shift: The marginal input feature distribution varies between source and target domains ($P_s(X) \neq P_t(X)$) while the posterior class mapping remains invariant ($P_s(Y \mid X) = P_t(Y \mid X)$). In agricultural imaging, covariate shift arises from variations in camera sensor optics, ambient illumination, background surfaces, capture geometry, and regional cultivar appearance differences.
- Label Shift: The prior class distribution shifts between domains ($P_s(Y) \neq P_t(Y)$) while the class-conditional input densities $P(X \mid Y)$ remain identical.
- Concept Shift: The conditional relationship between visual features and semantic class labels changes between environments ($P_s(Y \mid X) \neq P_t(Y \mid X)$), such that identical visual appearances correspond to different quality states.
- Open-Set Semantic Shift: The target domain contains semantic classes whose support lies outside the source taxonomy: $\text{supp}(P_t(Y)) \nsubseteq \text{supp}(P_s(Y))$.

In this taxonomy, the FruitVision treatment-labelled cohort represents images of known fruit species carrying a dataset-reported formalin-mixed condition. We do not classify this condition as concept shift or a new semantic class, nor do we conflate post-harvest treatment-labelled shifts with label-space expansion. Rather, it represents an empirical condition shift wherein visual manifestations of post-harvest immersion are assessed against a classifier trained on natural produce, without implying independent chemical residue detection.

In agricultural image analysis, it is also essential to distinguish between domain adaptation and cross-domain evaluation without adaptation:

- Domain Adaptation: Methods in this paradigm actively adjust model parameters, feature representations, or latent feature alignments between a known source domain $\mathcal{D}_s$ and a target domain $\mathcal{D}_t$ during training or test-time calibration. Unsupervised domain adaptation (UDA) utilizes unlabelled target-

domain images to minimize divergence measures (such as Maximum Mean Discrepancy or Correlation Alignment) or employs adversarial domain discriminators to project source and target features into a domain-invariant latent space.

- Cross-Domain Evaluation Without Adaptation: Cross-domain evaluation without adaptation provides a diagnostic assessment of model performance on selected external repositories. It does not, by itself, establish performance under physical field deployment. Instead, this protocol evaluates the direct, unadapted generalization of a pre-trained classifier when exposed to external photographic datasets without access to target-domain training data, fine-tuning steps, or parameter updates, benchmarking baseline failure modes when adaptation is unavailable.

A notable contribution in agricultural domain adaptation is the study by Wang et al. (2023) [21], titled "A cross-domain fruit classification method based on lightweight attention networks and unsupervised domain adaptation" (Complex & Intelligent Systems, 9(4):4227–4247). Wang et al. proposed HAM-MobileNet, integrating a hybrid spatial-channel attention module into MobileNetV3, coupled with an unsupervised domain adaptation framework that minimizes a domain discrepancy loss combining Local Maximum Mean Discrepancy (LMMD) and Correlation Alignment (CORAL). Their framework was evaluated across two fruit classification benchmarks: (1) a grape classification dataset containing seven grape varieties across four domains (controlled environment, illumination variation, background variation, and internet images); and (2) a general fruit classification dataset containing 11 fruit types across two domains. Their findings demonstrated that active domain alignment significantly improves target-domain classification performance relative to unadapted baselines.

While Wang et al. [21] provide valuable insights into active domain alignment, their methodology differs fundamentally from the present study. Wang et al. focus on model training and domain adaptation to maximize transfer performance. In contrast, the present investigation evaluates the diagnostic failure modes of a common frozen MobileNetV2 classifier without adapting its weights to the target repositories. Rather than modifying model parameters, our study investigates: (1) the magnitude of external known-class accuracy differences across distinct repository sources; and (2) whether post-hoc out-of-distribution detection methods can reliably identify and reject out-of-domain samples before misclassification occurs. Furthermore, as the underlying datasets, species definitions, backbones, and evaluation protocols differ, reported numerical accuracy values cannot be compared directly across incompatible study protocols.

2.3. Open-Set Recognition and Out-of-Distribution Detection

When deployed in unconstrained real-world environments, computer vision models inevitably encounter visual inputs that deviate from their training distributions. The inability of standard closed-set classifiers to handle such inputs motivated the formalization of Open-Set Recognition (OSR) and generalized Out-of-Distribution (OOD) detection [13–15].

Scheirer et al. (2013) [13] formalized open-set recognition as balancing empirical risk minimization on known training classes against open-space risk—the risk of erroneously assigning unknown inputs to known semantic classes. Bendale and Boult (2016) [14] extended this principle to deep neural networks, demonstrating that deep classifiers suffer from intrinsic overconfidence: deep networks map inputs into linear feature spaces capped by a softmax activation function $\sigma(z)_k = \exp(z_k)/\sum_j \exp(z_j)$, which forces posterior probabilities to sum to unity. Consequently, an unfamiliar, anomalous, or unseen-class input can produce near-maximal softmax confidence if its projection aligns even slightly closer to one class weight vector than others, even when the input resides far from all known training clusters in latent space [8, 9, 19].

Generalized out-of-distribution detection expands open-set recognition beyond semantic class novelty, addressing covariate domain shifts, environmental background variations, and corrupted sensor inputs [15]. In this study, we systematically evaluate five established post-hoc OOD scoring algorithms, applied to a common frozen MobileNetV2 classifier without architectural alteration:

1. Maximum Softmax Probability (MSP) [9]: Formulated by Hendrycks and Gimpel (2017) as an established confidence baseline, MSP utilizes the maximum predicted posterior probability as the confidence

score:

$$S_{\text{MSP}}(x) = \max_{k\in\{1,\dots,K\}} \frac{\exp(z_k(x))}{\sum_{j=1}^{K}\exp(z_j(x))} \tag{1}$$

where $z_k(x)$ represents the logit for class k. While computationally straightforward, MSP represents maximum softmax confidence scoring rather than calibrated probability estimation, and it frequently fails under severe distribution shift due to softmax probability saturation on out-of-distribution inputs.

2. OpenMax [14]: Introduced by Bendale and Boult (2016), OpenMax replaces the standard softmax layer by modeling the distribution of penultimate-layer activation vectors for each known class using Extreme Value Theory (EVT). Class-conditional mean activation vectors (MAVs) are computed on training data, and Weibull distributions are fitted to the largest distances between training samples and their respective MAVs. For an operational input x, OpenMax computes class-specific rejection probabilities, modifies activation magnitudes, and redistributes activation mass to an explicit "unknown" class ($P_0(x)$). An input is accepted only if $1 - P_0(x)$ exceeds a pre-calibrated threshold.
3. ODIN (Out-of-Distribution detector for Neural networks) [16]: Liang et al. (2018) demonstrated that OOD detection can be significantly enhanced without retraining through two post-hoc operations: temperature scaling ($T > 1$) and input-space gradient perturbation. Logits are scaled by temperature T, smoothing the probability distribution, while the input image x is perturbed via gradient ascent:

$$\tilde{x} = x - \epsilon\,\text{sign}\left(-\nabla_x \log \frac{\exp(z_{\hat{y}}(x)/T)}{\sum_{j=1}^{K}\exp(z_j(x)/T)}\right) \tag{2}$$

where $\hat{y} = \arg\max_k z_k(x)$. This directional perturbation increases the maximum softmax probability of in-distribution instances substantially more than that of out-of-distribution instances, amplifying the separation margin.

4. Mahalanobis Feature Distance [17]: Lee et al. (2018) proposed modeling penultimate feature representations $f(x)$ as a mixture of class-conditional multivariate Gaussians with class centroids μ_k and a tied covariance matrix $\mathbf{\Sigma} = \frac{1}{N}\sum_{k=1}^{K}\sum_{i:y_i=k}(f(x_i)-\mu_k)(f(x_i)-\mu_k)^\top$. The anomaly score is defined as the negative minimum Mahalanobis distance to the nearest class centroid:

$$S_{\text{Maha}}(x) = -\min_{k\in\{1,\dots,K\}} (f(x)-\mu_k)^\top \mathbf{\Sigma}^{-1}(f(x)-\mu_k). \tag{3}$$

This squared distance quantifies the statistical deviation of the latent feature representation under the fitted class-conditional Gaussian feature model. In empirical implementations, the negated Euclidean Mahalanobis distance $-\sqrt{(f(x)-\mu_k)^\top\mathbf{\Sigma}^{-1}(f(x)-\mu_k)}$ is frequently computed instead; because the square root is a strictly monotonic transformation on non-negative quadratic distances, it preserves sample ranking, AUROC, and directional orientation without altering thresholded decision properties.

5. Energy-Based Out-of-Distribution Detection [18]: Liu et al. (2020) demonstrated that mapping logits directly to Helmholtz free energy provides a theoretically grounded OOD scoring function that avoids the normalisation artifacts of softmax probabilities. The negative free energy is defined as:

$$S_{\text{Energy}}(x) = -E(x;f) = T\cdot\log\sum_{k=1}^{K}\exp\left(\frac{z_k(x)}{T}\right) \tag{4}$$

The energy score is derived from the log-sum-exp of the classifier logits and is motivated by an energy-based interpretation of the classifier [18]. In this study, it is used as an OOD ranking score, not as a calibrated estimate of the true input marginal density.

Methodological Positioning: These five algorithms represent distinct mathematical principles: maximum softmax confidence scoring (MSP), extreme value activation modeling (OpenMax), temperature and input gradient perturbation (ODIN), multivariate Gaussian feature distance (Mahalanobis), and unnormalized logit thermodynamics (Energy). None of these algorithms was specifically engineered for agricultural imagery; they were designed and benchmarked on general vision tasks (CIFAR, SVHN, ImageNet) [15]. In this study, we apply these established post-hoc scoring functions to a common frozen MobileNetV2 produce classifier across source-stratified fruit cohorts. Furthermore, the operational Energy–MSP Dual-Gate cascade investigated in our operational analysis is an empirical decision policy combining Energy and MSP at validation order-statistic operating thresholds (τ_{95}), and does not represent a new theoretical OOD scoring algorithm.

2.4. Open-Set Recognition in Food and Agricultural Applications

Recognizing the vulnerability of closed-set classifiers in unconstrained operational settings, recent studies have begun exploring open-set recognition and anomaly detection within agricultural and food-processing domains.

Sattar et al. (2024) — SensorNet [20]: Titled "A comprehensive approach to detecting chemical adulteration in fruits using computer vision, deep learning, and chemical sensors" (Intelligent Systems with Applications, 23:200402), this study investigated the detection of chemical adulteration (specifically formaldehyde immersion) in fruits. Sattar et al. formulated SensorNet, a multimodal framework integrating captured fruit imagery with chemical-sensor measurements from an electronic sensor array sensitive to formaldehyde gas. An essential methodological distinction exists between SensorNet and the present investigation: SensorNet is a multimodal sensor-fusion system that integrates optical images with direct chemical-sensor measurements. In contrast, our study evaluates image-level classification on photographic records carrying dataset-reported chemical treatment labels, with no independent chemical measurements conducted in the present benchmark. Because the sensing modalities and target outputs differ fundamentally, numerical classification accuracy in SensorNet cannot be directly compared to optical OOD AUROC or operational false acceptance rates.

Pehlivan and Kara (2026) [22]: Titled "A Lightweight One-Shot Open-Set Metric Learning Framework for Food Recognition and Decision Support in Smart Ovens" (Electronics, 15(16):3533), this study developed an open-set metric learning framework using a modified MobileNetV2 architecture for food recognition and unknown object rejection in smart kitchen appliances. By replacing the traditional softmax layer with normalized feature embeddings evaluated via cosine similarity against stored class centroids, their system rejects unknown food items below a calibrated similarity threshold. While Pehlivan and Kara utilize a MobileNetV2 backbone, their application domain focuses on one-shot recognition of prepared food dishes within automated domestic cooking cavities. In contrast, our study evaluates whole biological produce across multi-source external photographic repositories under natural post-harvest distribution shifts.

Jansi and Kavitha — PEOS-Net (2026) [23]: Titled "PEOS-Net: A Prototype-Energy Open-Set Network for Robust Tomato Leaf Disease Recognition Beyond the Closed-World Assumption" (Frontiers in Artificial Intelligence, Volume 9, Article 1938398; published October 30, 2026), this peer-reviewed study introduced a prototype-energy open-set network for crop leaf disease recognition using a frozen EfficientNet-B0 backbone. The architecture couples a prototype-based classification branch with an energy-based OOD scoring branch, evaluated on an eight-class tomato subset of the PlantVillage repository, where two tomato disease classes were held out as unknown categories. The study incorporates held-out calibration and test partitions, multiple random seeds, and statistical evaluations. An important limitation of the study is that the unknown classes were drawn from the identical PlantVillage repository, crop species, and imaging environment as the known training classes. Consequently, PEOS-Net represents a within-dataset class-holdout evaluation and does not establish external field-condition robustness. Furthermore, because foliar pathology

involves localized planar lesions on leaves rather than whole-fruit volumetric senescence, its task setting differs fundamentally from post-harvest produce grading.

Yan et al. — PAG-MSBNet (2026) [24]: Titled "PAG-MSBNet: Prototype-aware gated multi-scale branch network for open-set crop disease recognition" (Ecological Informatics, Volume 95, Article 103703, 2026), this study formulated a multi-scale spatial-channel branch network incorporating prototype-aware gating mechanisms to classify known crop diseases while rejecting novel pathologies across public plant disease benchmarks, including PlantVillage and Paddy Doctor. Yan et al. focus on custom architectural engineering for open-set crop leaf diagnostics. In contrast, our study evaluates a standard, frozen commercial vision backbone (MobileNetV2) across external photographic repositories of harvested fruits, quantifying empirical failure modes under source-stratified covariate, species, and post-harvest treatment shifts without modifying the underlying model architecture.

2.5. Critical Literature Comparison and Research Gap

To synthesize the methodological landscape and clarify the precise scientific positioning of this study, Table 1 provides a structured comparison of recent representative studies across agricultural computer vision, domain adaptation, and open-set recognition.

2.5.1. Critical Synthesis and Specific Research Gap

The studies reviewed address complementary problems in fruit-quality classification, domain adaptation, chemical-sensor-assisted adulteration detection, and open-set recognition. The present study examines a particular combination of evaluation dimensions: established post-hoc OOD scores applied to a common frozen fruit-quality classifier, evaluated across external known-class, unseen-species, treatment-labelled, and combined-shift cohorts, with source-matched comparisons and fixed validation-calibrated operational thresholds.

Specifically, the present study investigates four evaluation dimensions:

1. Supervised Closed-Set Classification vs. External OOD Evaluation: Existing fruit-freshness studies predominantly report supervised classification performance under random train/test partitions within single curated datasets [4–6, 10]. Such evaluations do not assess how classifiers behave when exposed to external repositories or out-of-distribution instances. Our study addresses this gap by assessing a frozen classifier on external multi-repository cohorts and evaluating post-hoc OOD detection.
2. Source-Matched Cohort Comparisons to Mitigate Cross-Repository Confounding: In open-set evaluations where out-of-distribution cohorts are drawn from an external repository distinct from the training set, differences in camera hardware, ambient illumination, and background context present a substantial methodological risk of being confounded with true semantic divergence [15]. Source-matched comparisons are conducted within the same repository to reduce repository-level confounding, though potential within-repository capture variations may remain.
3. Empirical Evaluation of Treatment-Labelled Images: While multimodal systems (such as SensorNet [20]) incorporate specialized chemical gas sensors to identify adulteration, our study evaluates whether standard RGB imagery contains sufficient optical separation signal for post-hoc OOD scoring methods applied to a standard frozen classifier to distinguish treatment-labelled images (documented under a dataset-reported 10% formalin immersion protocol [12]) from natural produce, without claiming independent chemical detection.
4. Distinguishing Retrospective Ranking from Operational Threshold Calibration: Many OOD detection studies prioritize retrospective metrics: AUROC assesses ranking discrimination across all possible decision thresholds without selecting an operating point, whereas FPR@95 evaluates the false positive rate at a retrospective threshold chosen post-hoc to achieve 95% true positive rate on the test cohort [9, 15, 18]. While suitable for relative algorithm comparison, retrospective metrics do not reflect operational deployment, where a discrete acceptance threshold must be pre-calibrated on validation data and held fixed during external evaluation. Measuring operational Unknown False Acceptance Rates

Table 1. Structured literature comparison of representative agricultural computer vision, domain adaptation, and open-set recognition studies alongside the present investigation. This table reflects a literature-based comparison of reported study characteristics; it does not represent a direct quantitative head-to-head evaluation across incompatible datasets and evaluation protocols.

Study & Venue	Application Domain	Input Data & Modality	Core Methodology	Distinction from Present Study
Sultana et al. (2022) [11] Data in Brief	Fruit quality dataset curation	RGB images (16 fresh/rotten classes across 8 fruit types)	Photographic dataset collection from retail markets and fields across 16 fresh/rotten classes (8 fruit types)	Dataset descriptor publication. Evaluated as an external benchmark cohort in our study ($N = 3,193$ audited image records).
Bijoy et al. (2025) [12] Data in Brief	Fruit quality and adulteration dataset	RGB images (Fresh, rotten, formalin-mixed across 5 fruits)	Photographic dataset collection across 3 mobile smartphone cameras; augmentations	Dataset descriptor publication. Evaluated as our primary multi-shift benchmark repository ($N = 10,045$ audited image records).
Abayomi-Alli et al. (2024) [10] Multimed. Tools Appl.	Fruit freshness dataset curation	RGB images (11 fruit types; Fresh, Mild, Rotten)	Evaluates 5 baseline deep architectures (ShuffleNet, SqueezeNet, EfficientNet, ResNet18, MobileNetV2)	Multi-fruit freshness benchmark. Evaluated under closed-set classification without post-hoc OOD detection or external domain shift analysis.
Wang et al. (2023) [21] Complex & Intell. Syst.	Cross-domain fruit classification	RGB images (Grape 4-domain set; 11-fruit 2-domain set)	HAM-MobileNet (MobileNetV3 + hybrid attention) + Unsupervised Domain Adaptation (LMMD + CORAL)	Focuses on active model retraining and domain alignment. Present study evaluates a frozen classifier and post-hoc OOD detectors.
Sattar et al. (2024) [20] Intell. Syst. with Appl.	Fruit chemical adulteration detection	Multimodal: RGB images + electronic chemical gas sensors	SensorNet: hybrid CNN visual feature extraction coupled with chemical sensor array	Multimodal hardware framework integrating chemical gas sensors. Present study investigates optical RGB vision limits under treatment labels.
Pehlivan & Kara (2026) [22] Electronics	Food recognition in smart kitchen appliances	RGB images (Prepared food dataset)	Modified MobileNetV2 with open-set metric learning (cosine similarity thresholding)	Evaluates prepared food dishes in domestic cooking cavities. Present study evaluates whole fruit spoilage across external photographic repositories.
Jansi & Kavitha (2026) [23] Frontiers in Artif. Intell.	Open-set crop disease recognition	RGB images (PlantVillage tomato leaf dataset)	PEOS-Net: Prototype learning + energy-based OSR on frozen EfficientNet-B0; held-out classes	Within-dataset class-holdout evaluation on foliar lesions. Present study evaluates whole post-harvest fruit senescence across external repositories.
Yan et al. (2026) [24] Ecological Informatics	Open-set crop disease recognition	RGB images (Paddy Doctor, PlantVillage crop leaves)	PAG-MSBNet: Prototype-aware gating + multi-scale spatial-channel branch network	Architectural engineering for crop leaf pathology. Present study evaluates established post-hoc OOD scorers on a frozen standard backbone.
Present study (V4.1 evaluation)	Diagnostic evaluation of post-harvest fruit vision	Audited RGB image census ($N = 13,465$ image records; Primary, FruitVision, Sultana)	Diagnostic benchmark: 5 post-hoc OOD detectors (MSP, Energy, ODIN, Maha, OpenMax) on frozen MobileNetV2; Dual-Gate cascade	Compares source-matched cohorts to reduce repository confounding under covariate, species, and treatment shifts; bootstrap CIs; pHash auditing.

(UFAR) alongside Known-Class Rejection Rates (KRR) under fixed validation thresholds provides complementary, deployment-context measurements of operational trade-offs, supported by 1,000-iteration

non-parametric bootstrap confidence intervals and perceptual-hash (pHash) near-duplicate sensitivity analysis across an audited 13,465-record image census.

3. Dataset Census and Experimental Design

This section details the source-stratified evaluation benchmark, comprising $N = 13,465$ audited photographic image records assembled from three independent repositories. We present the study design rationale, repository provenance, label-mapping protocol, five-partition census, four source-matched evaluation cohorts, duplicate and perceptual-hash sensitivity analysis, experimental separation safeguards, and reproducibility artifacts.

3.1. Study Design Overview

Evaluating automated fruit-quality classification models under practical deployment conditions presents a complex experimental challenge. When vision models trained on curated image datasets are evaluated on external imagery, observed performance degradation typically reflects multiple confounding factors acting simultaneously: variations in camera sensor optics, ambient illumination, photographic backgrounds, biological cultivars, and post-harvest handling marks [10–12]. If images from diverse sources are simply pooled into an aggregated test set, it becomes impossible to determine whether detection failures stem from semantic novelty (encountering an unseen fruit species), physiological or surface anomalies (encountering damaged or chemically treated produce), or mere covariate shift in image acquisition conditions.

To resolve these compounding factors, this study implements a source-stratified empirical evaluation. We freeze a pre-trained six-class MobileNetV2 classifier [7] and evaluate it across structured image cohorts drawn from three independent repositories: the Primary Kaggle dataset, FruitVision [12], and Sultana et al. [11]. Crucially, the experimental architecture maintains a strict distinction between:

1. Closed-Set Generalization (Partitions A and B): Evaluating accuracy, balanced accuracy, and class-wise confusion on known classes across laboratory baseline conditions (Partition A, $N = 227$) versus external retail and field environments (Partition B, $N = 5,350$), holding the label space $\mathcal{Y} = \{0, 1, \ldots, 5\}$ constant.
2. Out-of-Distribution Detection (Partitions C, D1, and D2): Evaluating the capacity of post-hoc scoring functions to detect and reject non-target inputs, including treatment-labelled condition shifts (Partition C, $N = 1,917$), unseen fruit species (Partition D1, $N = 4,776$), and compound shifts (Partition D2, $N = 1,195$).

Furthermore, to separate the specific mechanism of out-of-distribution (OOD) shift from cross-repository acquisition bias, we structure the OOD evaluation into four source-matched cohorts (Cohorts A, B, C, and D). By comparing in-distribution known images and out-of-distribution anomalous images collected within the same parent repository, source matching reduces repository-level provenance differences, while within-repository acquisition variation remains possible. Throughout this study, the evaluation unit is designated strictly as an image record; photographic records cannot be assumed to represent statistically independent biological specimens.

3.2. Dataset Sources and Provenance

The benchmark image census is compiled from three independent photographic archives, each characterized by distinct acquisition settings, camera modalities, and environmental conditions. Table 2 summarizes the provenance, published composition, evaluated subsets, and key characteristics of each repository.

3.2.1. Primary Kaggle Dataset

The primary training and baseline evaluation partition originates from a widely utilized curated benchmark repository for fruit freshness classification. The repository covers three commercial fruit types—apple

Table 2. Dataset source and provenance summary. Evaluated record counts represent audited, unique image records included in the present study after hash-identified duplicate removal, distinguishing local benchmark subsets from published repository totals.

Source Repository	Citation / Venue	Published Composition	Evaluated N	Evaluated Species & Conditions	Capture Environment & Key Provenance
Primary Kaggle Archive	Curated public benchmark	2,043 test images (227 raw, 1,816 aug.)	227	Apple, banana, orange (Fresh, Rotten)	Controlled laboratory-style conditions with uniform backgrounds; neutral illumination; baseline ID test set (Partition A).
FruitVision	Bijoy et al. (2025) [12] Data in Brief	10,154 raw images (81,232 aug. archive)	10,045	Apple, banana, orange, grape, mango (Fresh, Rotten, Formalin-mixed)	3 mobile smartphone cameras (Samsung A50, Redmi Note 7, iPhone 11); indoor/outdoor retail lighting; 109 hash-identified duplicates excluded.
Sultana et al.	Sultana et al. (2022) [11] Data in Brief	16 classes, 8 fruits (~3,200 raw images)	3,193	Apple, banana, orange, grape, guava, jujube, pomegranate, strawberry (Fresh, Rotten)	Retail fruit shops and open agricultural fields; natural ambient illumination; agricultural specialist annotation; 7 hash-identified duplicates excluded.
Total Benchmark	Three Repositories	Multiple Archives	13,465	8 Fruit Species, 3 Conditions	Audited multi-source evaluation census (Partitions A–D2).

(Malus domestica), banana (Musa acuminata), and orange (Citrus sinensis)—each categorized into fresh and rotten states. Photography was conducted under controlled laboratory-style conditions with relatively uniform, neutral backgrounds and stable illumination. The published test partition contains 2,043 files consisting of 227 raw photographic images and 1,816 synthetic geometric and photometric augmentations. In this study, we strictly isolate the 227 raw, unaugmented photographic images ('is_raw == 1') to construct Partition A (Primary In-Distribution Test). The 1,816 synthetically augmented images were set aside to restrict evaluation strictly to authentic photographic captures; these represent upstream synthetic transformations and are not classified as duplicate removals. An independent validation partition of 227 raw images was maintained strictly for operating-threshold calibration (see Section 3.7).

3.2.2. FruitVision Repository

The FruitVision dataset was introduced by Bijoy et al. [12] as a benchmark for produce freshness, rot, and adulteration recognition. The repository encompasses five fruit species: three known species (apple, banana, orange) and two non-target species (grape Vitis vinifera and mango Mangifera indica). Photography was conducted across diverse wholesale markets, retail stalls, and laboratory benches using three commercial smartphone devices: Samsung Galaxy A50 (25 MP primary sensor), Xiaomi Redmi Note 7 (48 MP primary sensor), and Apple iPhone 11 (12 MP primary sensor). Image capture spanned varied ambient indoor lighting, direct outdoor sunlight, and fluorescent retail illumination.

A distinctive feature of FruitVision is its inclusion of a dataset-reported treatment protocol: fruits were immersed in a solution of 10% formalin and 90% water for 30 minutes, allowed to air-dry, and photographed 48 hours post-immersion under natural ambient conditions alongside untreated fresh and naturally decaying specimens. While Bijoy et al. published 10,154 raw photographs (expanded via photometric and geometric transformations to an 81,232-image augmented archive), the present study extracts exclusively the raw photographic collection. Cryptographic hash auditing identified and removed 109 hash-identified duplicate files, yielding exactly 10,045 audited image records evaluated across Partitions B, C, D1, and D2. We emphasize that treatment-labelled images are evaluated strictly under their dataset-reported labels; no independent chemical assays or residue validations were conducted.

3.2.3. Sultana et al. Repository

The dataset introduced by Sultana et al. [11] was curated to recognize produce quality across authentic retail and farm environments. The dataset contains 16 fresh and rotten categories representing eight distinct fruit species: apple, banana, orange, grape, guava (Psidium guajava), jujube (Ziziphus jujuba), pomegranate (Punica granatum), and strawberry (Fragaria × ananassa). Images were captured in local retail shops and active agricultural orchards under ambient natural lighting, with fruit specimen selection and freshness categorization verified with assistance from an agricultural domain specialist. We harvested 3,200 raw photographic records (200 records per class across all 16 classes). MD5 cryptographic hashing identified 7 hash-identified duplicate files within the orange fresh class, leaving exactly 3,193 audited image records evaluated across Partitions B and D1.

3.3. Known-Class Definition and Label Mapping

The classification space of the frozen MobileNetV2 backbone is defined over $K = 6$ discrete semantic classes:

$$\begin{aligned}\mathcal{Y} = \{&0 : \text{fresh_apple},\ 1 : \text{fresh_banana},\ 2 : \text{fresh_orange},\\ &3 : \text{rotten_apple},\ 4 : \text{rotten_banana},\ 5 : \text{rotten_orange}\}.\end{aligned} \tag{5}$$

To integrate external imagery into this framework, we apply unambiguous label-mapping rules governed by the underlying biological species and physical freshness condition:

1. External Known-Class Mapping (Partition B): External images from FruitVision and Sultana et al. depicting apple, banana, or orange in natural fresh or rotten states are mapped directly to their corresponding index in $\mathcal{Y}$. For example, FruitVision Fresh_Apple maps to class 0, and Sultana rotten_orange maps to class 5. No class collapsing across fresh/rotten boundaries or cross-species reassignments are permitted.
2. Unseen-Species OOD Mapping (Partition D1): Images depicting botanical species outside the set $\{\text{apple}, \text{banana}, \text{orange}\}$ in natural fresh or rotten conditions are designated as out-of-distribution unseen species. This includes grape and mango from FruitVision, and grape, guava, jujube, pomegranate, and strawberry from Sultana et al.
3. Treatment-Labelled Condition Shift Mapping (Partition C): Images from FruitVision depicting known species (apple, banana, orange) photographed under the dataset-reported formalin-immersion condition are designated as treatment-labelled condition shifts. Because the classifier was trained exclusively on untreated produce, these images evaluate whether post-hoc OOD scoring functions discriminate images bearing the dataset-reported treatment label from untreated knowns. This evaluation tests model behavior on produce labelled under altered post-harvest handling; it does not claim independent chemical detection, residue verification, or the establishment of treatment-caused visual anomalies.
4. Combined Shift Mapping (Partition D2): Images from FruitVision depicting unseen species (grape and mango) subjected to the dataset-reported formalin-immersion condition are designated as combined shifts, presenting simultaneous semantic and treatment condition divergence.

No subjective manual pruning or post-hoc relabeling was performed; all annotations adhere strictly to the verified manifests of the respective source publications.

3.4. Image-Level Census and Partition Construction

The complete evaluation benchmark comprises $N = 13,465$ audited photographic image records structured into five mutually exclusive partitions, designated Partitions A through D2. Table 3 presents the detailed

Table 3. Evaluation partition census ($N = 13,465$). Partitions are mutually exclusive based on unique SHA-256 and MD5 hashes. Counts represent individual photographic image records rather than independent biological specimens.

Partition	Evaluation Role	Source Repository	Target Status	Species, Conditions, and Sub-counts	Partition N
Partition A	Primary In-Distribution (ID Test)	Primary Kaggle	Known ($\mathcal{Y}$)	Apple (F 34, R 49), Banana (F 33, R 47), Orange (F 31, R 33)	227
Partition B	External Known-Class Shift	FruitVision Sultana et al.	Known ($\mathcal{Y}$)	FruitVision ($N = 4,157$): Apple (F 765, R 630), Banana (F 748, R 631), Orange (F 753, R 630) Sultana ($N = 1,193$): Apple (F 200, R 200), Banana (F 200, R 200), Orange (F 193, R 200)	5,350
Partition C	Treatment-Labelled Condition Shift	FruitVision	Unknown (OOD Condition)	Formalin-mixed Knowns ($N = 1,917$): Apple (641), Banana (660), Orange (616)	1,917
Partition D1	Unseen-Species Pure Semantic Shift	FruitVision Sultana et al.	Unknown (OOD Species)	FruitVision ($N = 2,776$): Grape (F 753, R 630), Mango (F 763, R 630) Sultana ($N = 2,000$): Grape (F 200, R 200), Guava (F 200, R 200), Jujube (F 200, R 200), Pomegranate (F 200, R 200), Strawberry (F 200, R 200)	4,776
Partition D2	Combined Semantic & Treatment Shift	FruitVision	Unknown (OOD Compound)	Formalin-mixed Unseen ($N = 1,195$): Grape (587), Mango (608)	1,195
Total	Complete Evaluation Benchmark	Three Repositories	Audited Census	5 Mutually Exclusive Partitions ($A + B + C + D1 + D2$)	13,465

census, detailing the evaluation role, source repositories, target classification status, species composition, and exact record counts.

The census reconciles precisely across source repositories:

- Primary Kaggle: Contributes 227 image records, all within Partition A.
- FruitVision: Contributes 10,045 image records, spanning Partition B ($N = 4,157$), Partition C ($N = 1,917$), Partition D1 ($N = 2,776$), and Partition D2 ($N = 1,195$).
- Sultana et al.: Contributes 3,193 image records, spanning Partition B ($N = 1,193$) and Partition D1 ($N = 2,000$).

Summing across sources gives $227 + 10,045 + 3,193 = 13,465$ records. Similarly, summing across partitions yields $227 + 5,350 + 1,917 + 4,776 + 1,195 = 13,465$ records. Cryptographic hash analysis confirms that all 13,465 images possess unique SHA-256 and MD5 checksums, verifying complete mutual exclusivity.

3.5. Source-Stratified Evaluation Cohorts

To evaluate out-of-distribution detection without conflating produce anomalies with cross-dataset domain shifts, we establish four source-matched evaluation cohorts, summarized in Table 4. In standard unstratified evaluations, pairing in-distribution images from one repository (e.g., Kaggle) against out-of-distribution images from another (e.g., FruitVision) introduces severe macro-confounders: differences in background chromaticity, ambient illumination, and sensor colour calibration enable detectors to distinguish sources rather than semantic or condition anomalies.

The four source-matched cohorts isolate specific distributional dimensions:

Table 4. Source-matched out-of-distribution evaluation cohorts (Cohorts A–D). Source matching reduces repository-level provenance differences, while within-repository acquisition variation remains possible.

Cohort	Known Reference (N_{id})	Shifted / OOD (N_{ood})	Source Repository	Shift Dimension	Primary Research Question
Cohort A	FruitVision Known (Partition B, $N = 4,157$)	FruitVision Unseen (Partition D1, $N = 2,776$)	FruitVision	Within-Repository Semantic Shift	Can post-hoc detectors identify unseen fruit species (grape, mango) within the same repository collection?
Cohort B	FruitVision Known (Partition B, $N = 4,157$)	FruitVision Formalin-mixed (Partition C, $N = 1,917$)	FruitVision	Within-Repository Treatment Shift	Can post-hoc detectors or frozen classifiers discriminate dataset-reported treatment-labelled condition shifts on familiar fruit species?
Cohort C	FruitVision Known (Partition B, $N = 4,157$)	FruitVision Preserved Unseen (Partition D2, $N = 1,195$)	FruitVision	Within-Repository Combined Shift	How does compound shift (unseen species bearing dataset-reported treatment labels) affect detector discrimination?
Cohort D	Sultana Known (Partition B, $N = 1,193$)	Sultana Unseen (Partition D1, $N = 2,000$)	Sultana et al.	Within-Repository Semantic Shift	How does unseen-species detection generalize across field and retail market imagery containing diverse unseen agricultural species?

1. Cohort A (Within-FruitVision Semantic Shift): Compares $N_{\text{id}} = 4,157$ natural known images (apple, banana, orange from Partition B) against $N_{\text{ood}} = 2,776$ natural unseen-species images (grape, mango from Partition D1). As both subsets originate from the FruitVision repository, this comparison reduces cross-repository acquisition bias, though within-repository capture variations remain possible.
2. Cohort B (Within-FruitVision Treatment Condition Shift): Compares the same $N_{\text{id}} = 4,157$ natural known images against $N_{\text{ood}} = 1,917$ images of the identical three fruit species bearing the dataset-reported treatment-labelled condition (Partition C). This cohort assesses detector behavior on images labelled under post-harvest chemical immersion protocols, without asserting independent chemical verification or treatment-induced visual artifacts.
3. Cohort C (Within-FruitVision Combined Shift): Compares $N_{\text{id}} = 4,157$ natural known images against $N_{\text{ood}} = 1,195$ images of unseen species bearing the dataset-reported treatment-labelled condition (grape and mango from Partition D2), evaluating the empirical interaction between semantic and treatment condition labelling.
4. Cohort D (Within-Sultana Semantic Shift): Compares $N_{\text{id}} = 1,193$ natural known images from Sultana et al. against $N_{\text{ood}} = 2,000$ natural unseen-species images (grape, guava, jujube, pomegranate, strawberry from Partition D1). This cohort evaluates semantic shift under outdoor farm and retail market conditions across diverse unseen agricultural species.

Importantly, Cohorts A, B, and C all utilize FruitVision Known (Partition B, $N_{\text{id}} = 4,157$) as their common in-distribution reference subset, evaluating distinct shifted cohorts (unseen species in Partition D1, treatment condition in Partition C, and combined shift in Partition D2, respectively) against this shared baseline. Because these source-matched cohorts reuse the same underlying reference records, they do not constitute statistically independent datasets. Rather, they provide comparative diagnostic probes of different distribution shift mechanisms against a unified, fixed reference distribution within the FruitVision repository.

We note an important methodological qualification: source matching substantially reduces repository-level provenance differences, but cannot eliminate potential micro-variations in capture angle, background clutter, or fruit specimen batching within a given repository.

3.6. Image Curation, Duplicate Handling, and Sensitivity Analysis

Photographic archives in agricultural computer vision frequently contain near-duplicate captures resulting from camera burst shooting, minor viewpoint adjustments, or synthetic data augmentation [10, 11]. To guarantee experimental integrity, we implemented a two-stage deduplication and sensitivity auditing protocol:

3.6.1. Hash-Identified Duplicate Auditing

Every candidate image file was audited via MD5 and SHA-256 cryptographic hashing. Duplicate files identified through matching cryptographic digests were purged prior to partition assignment. This forensic audit identified and removed 109 hash-identified duplicate files from FruitVision:

- Partition B: 28 duplicates excluded (1 fresh banana, 1 rotten banana, 26 rotten orange).
- Partition C: 33 duplicates excluded (2 apple formalin-mixed, 31 orange formalin-mixed).
- Partition D1: 17 duplicates excluded (17 grape fresh).
- Partition D2: 31 duplicates excluded (23 grape formalin-mixed, 8 mango formalin-mixed).

From Sultana et al., 7 hash-identified duplicate files were detected and purged from Partition B (7 orange fresh). In total, 116 hash-identified duplicate files were excluded across evaluated external repositories, ensuring that all 13,465 evaluated records represent unique files.

We strictly distinguish these 116 hash-identified duplicate exclusions from the primary dataset curation protocol: in the primary Kaggle test split, 1,816 synthetically augmented images were set aside to isolate the 227 unaugmented raw photographic records (‘is_raw == 1’) for Partition A. These 1,816 exclusions represent the intentional omission of upstream synthetic augmentations, and are never classified as duplicate removals.

3.6.2. Perceptual Hash (pHash) Sensitivity Analysis

Beyond exact duplicate files, photographic archives may contain visually near-identical frames captured seconds apart during burst photography or camera repositions. To quantify whether near-duplicate frames artificially distort OOD detection metrics, we computed 64-bit discrete cosine transform (DCT) perceptual hashes (pHash) for all 13,465 images. Pairwise collision scans were conducted across Hamming distance radii $r \in \{0, 4, 8\}$, corresponding to:

- $r = 0$: exact perceptual hash match;
- $r \leq 4$: strict near-duplicate match;
- $r \leq 8$: permissive visual collision.

Crucially, these pairwise Hamming radii r are scanning thresholds, which are distinguished from the resulting filtered evaluation tiers. Based on detected collisions at $r \leq 8$, we constructed three nested evaluation tiers:

1. Primary Cohort ($N = 13,465$): The canonical evaluation benchmark, containing all unique image records following exact MD5 deduplication.
2. Tier 1 Sensitivity ($N = 13,409$): Excludes image records exhibiting cross-partition pHash collisions at $r \leq 8$ (instances where an image in an in-distribution partition closely resembles an image in an out-of-distribution partition). Because FruitVision Known (Partition B, $N_{\text{id}} = 4,157$) serves as the shared reference subset across Cohorts A, B, and C, cohort-level pHash exclusions are not additive; naively summing exclusions across individual cohorts repeatedly counts the same reference images. In Tier 1, exactly 11 unique reference records from Partition B are excluded across all three FruitVision cohorts, alongside 14 shifted records from Partition D1 (Cohort A), 11 shifted records from Partition

C (Cohort B), and 20 shifted records from Partition D2 (Cohort C), with zero cross-partition collisions in Sultana et al. This yields a global benchmark total of exactly 56 unique excluded image records ($13,465 - 56 = 13,409$).

3. Tier 2 Sensitivity ($N = 13,382$): Further excludes image records exhibiting intra-partition multi-label collisions (near-identical images within the same partition bearing conflicting condition or sub-class labels). Tier 2 removes an additional 6 unique records from Partition B, 5 from Partition C, and 16 from Partition D1 (with 0 in Partition D2 and Sultana et al.), yielding 27 additional multi-label exclusions and an overall global benchmark total of exactly 83 unique excluded image records ($13,465 - 83 = 13,382$).

The empirical evaluation of detector performance across these three sensitivity tiers is presented in Section 5 (Sensitivity and Near-Duplicate Robustness Analysis), where metric stability is examined across scoring functions and cohorts.

Methodological Limitations of Perceptual Hashing: While this sensitivity framework provides a structured probe of metric stability under detected near-duplicate collisions, perceptual hashing has inherent limitations. Operating on low-frequency DCT coefficients, pHash cannot guarantee the detection of heavily cropped, rotated, or lighting-altered captures of the same physical fruit specimen, nor can it prevent collisions between distinct produce specimens with uniform surface colouration. Consequently, pHash sensitivity analysis provides a bounded robustness check against detectable visual collisions; it does not claim the complete elimination of all near-duplicates, burst-shooting captures, or specimen-level correlations in the underlying archives.

3.7. Evaluation Integrity and Experimental Separation

To maintain rigorous scientific standards and prevent methodological leakage, the experimental protocol enforces four strict integrity safeguards:

1. Separation of Validation Calibration and Evaluation Census: Prospective operating thresholds (τ_{95}, targeting approximately 95% in-distribution acceptance) were calibrated strictly on an independent primary validation partition ($N = 227$ raw Kaggle validation images, 'is_raw == 1'). Partition A comprises $N = 227$ raw Kaggle test images ('is_raw == 1'). Cryptographic auditing confirms that MD5 hash overlap between the reported validation calibration partition and Partition A is strictly zero, establishing that there is no exact-file overlap between the two sets. Furthermore, the 227 validation images are used exclusively for threshold estimation and are not included in the $N = 13,465$ evaluation census. While this cryptographic separation demonstrates the absence of exact duplicate files between calibration and test data, we emphasize that exact-hash verification is evidence-bounded: it establishes no file overlap, but does not prove the absence of every conceivable upstream correlation (such as potential near-duplicate specimens or sequential camera bursts in the underlying Kaggle collection).
2. A Priori Threshold Locking: All decision thresholds (e.g., $\tau_{\text{Energy}} = 1.054579$, $\tau_{\text{MSP}} = 0.700580$, and the Dual-Gate pair $\tau_E = 0.918451, \tau_M = 0.582610$) were locked on the validation partition using empirical order statistics and held fixed throughout all external evaluations. No threshold tuning, post-hoc threshold selection, or parameter optimization was performed on external partitions or test cohorts.
3. Model Checkpoint Immutability: The evaluated MobileNetV2 architecture (checkpoint MODEL_B_V4_1_CHECKPOINT.keras, with SHA-256 hash 4a12f354...b6b4) remained strictly frozen throughout all experiments. No transfer learning, fine-tuning, domain adaptation, or parameter updates were applied to any external cohort.
4. Physical and Chemical Non-Claim Boundary: This study evaluates digital RGB images under their published labels. It does not conduct chemical residue assays, does not verify biological specimen independence, and does not evaluate physical sorting machinery or live post-harvest conveyor operations.

3.8. Experimental Traceability and Data Availability

Experimental traceability is supported by structured digital artifacts maintained within the project research repository. We distinguish preserved internal artifacts from external runtime and source-access dependencies:

- Preserved Research Artifacts: Model weights, prediction logs, partition manifests, and calibration tables are preserved for inspection:
  - Image-level prediction records: STAGE4_1_IMAGE_LEVEL_PREDICTIONS.csv
  - Partition manifest and exclusion audit: STAGE4_0_3_CORRECTED_PARTITION_CENSUS.csv
  - Empirical metric point estimates: STAGE4_1_2_SOURCE_STRATIFIED_METRICS.csv
  - Non-parametric bootstrap confidence intervals: STAGE4_1_2_1_SOURCE_STRATIFIED_CONFIDENCE_INTERVALS.csv
  - Perceptual hash sensitivity evaluation tiers: STAGE4_1_2_1_SOURCE_STRATIFIED_PHASH_SENSITIVITY.csv
  - Verified claims ledger: STAGE4_1_2_1_1_FINAL_CLAIMS_LEDGER.csv
- External Dependencies: Full end-to-end execution of evaluation scripts depends on specific Python virtual environments, scientific machine-learning libraries (TensorFlow/Keras, scikit-learn, NumPy), underlying hardware drivers, and external network access to the original photographic archives hosted by third-party publisher repositories.

External photographic datasets utilized in this study were released by their respective authors under academic open-access licenses (e.g., Creative Commons Attribution 4.0 International for FruitVision [12] and Sultana et al. [11] via Data in Brief). In compliance with third-party licensing terms, raw photographic files must be accessed through their primary publisher archives, while all derived metadata, prediction matrices, and evaluation scripts are preserved for academic verification.

4. Methodology

This section details the computational methods, feature representations, post-hoc scoring algorithms, operational decision logic, threshold calibration protocol, and statistical reporting standards employed in this study. The present research constitutes an empirical evaluation and failure diagnosis of established methods rather than a proposal for a novel out-of-distribution (OOD) algorithm.

4.1. Methodological Overview

The end-to-end evaluation pipeline is illustrated in Fig. 1. The computational framework separates image processing and evaluation into four functional stages:

1. Image Ingestion and Preprocessing: Input photographic image records are decoded, bilinearly resized to $224 \times 224 \times 3$, and mapped to the standardized dynamic range $[-1.0, 1.0]$.
2. Frozen Backbone Inference and Feature Extraction: Preprocessed images are passed through a frozen MobileNetV2 architecture [7]. We extract both the 1280-dimensional penultimate feature representation $h(x) \in \mathbb{R}^{1280}$ (from global average pooling) and the unnormalized logit vector $z(x) \in \mathbb{R}^6$, from which softmax class posterior probabilities $p(x) = \mathrm{softmax}(z(x))$ are generated.
3. Post-hoc OOD Scoring: Five post-hoc scoring functions are computed from the extracted representations: Maximum Softmax Probability (MSP) [9], Energy-based scoring [18], an ODIN-style perturbation variant [16], Ledoit–Wolf Mahalanobis distance [17], and a modified OpenMax-style heuristic [14].
4. Operational Decision Gating and Multi-Level Evaluation: Continuous scores are mapped to discrete acceptance/rejection decisions via fixed prospective operating thresholds calibrated on validation data. Model performance is comprehensively evaluated across: (i) closed-set classification accuracy on

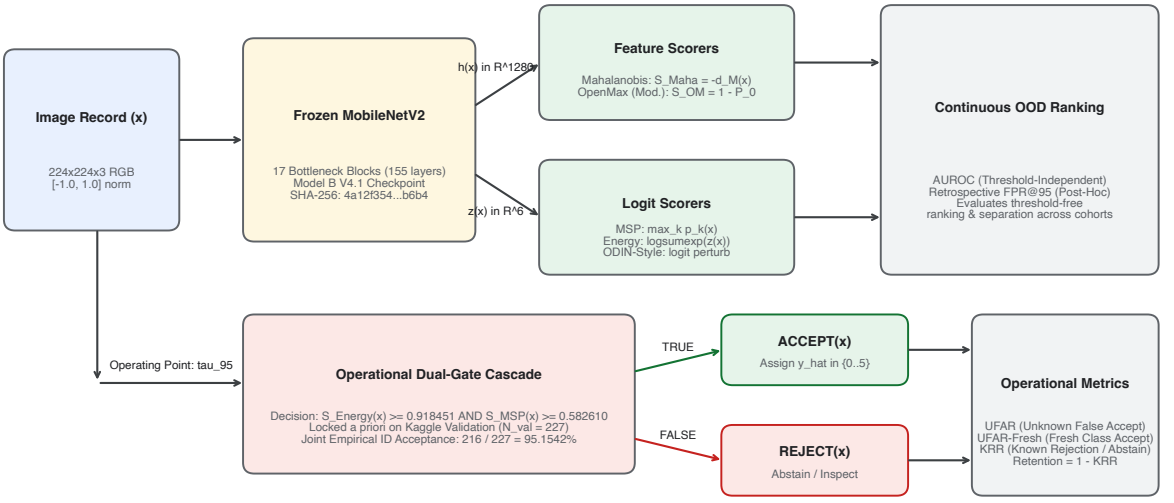


Fig. 1. End-to-end methodological pipeline of the source-stratified evaluation benchmark. Photographic image records are processed by a frozen MobileNetV2 backbone (checkpoint SHA-256: 4a12f354...b6b4) to extract penultimate features $h(x) \in \mathbb{R}^{1280}$ and logits $z(x) \in \mathbb{R}^6$. Five post-hoc scoring methods (MSP, Energy, an ODIN-style perturbation variant, Ledoit–Wolf Mahalanobis, and a modified OpenMax-style heuristic) are evaluated for OOD ranking and thresholded separation. Operating thresholds locked a priori on validation knowns ($N_{\text{val}} = 227$) govern discrete acceptance ($\hat{y} \in \{0, \ldots, 5\}$) or rejection ($\hat{y} = \text{Ⓡ}$) via the Energy–MSP Dual-Gate cascade.

knowns, (ii) threshold-independent OOD ranking (AUROC), (iii) retrospective operating-point separation (FPR@95), and (iv) fixed validation-threshold operational rejection metrics (UFAR, UFAR-Fresh, KRR).

4.2. Frozen MobileNetV2 Classifier

The classification backbone is a lightweight convolutional neural network based on the MobileNetV2 architecture [7], designed for resource-constrained edge execution.

4.2.1. Architecture and Output Space

The network comprises an initial 3×3 standard convolution followed by 17 bottleneck blocks: 1 initial inverted bottleneck stage without expansion and 16 inverted residual bottleneck blocks with depthwise separable convolutions and linear bottlenecks (expansion factor 6) [7]. In the Keras functional implementation, the feature extractor encompasses 155 layer operations terminating in a global average pooling layer that compresses spatial feature maps into a $D = 1280$ dimensional embedding $h(x) \in \mathbb{R}^{1280}$. The classification head consists of a dropout layer ($p = 0.3$) followed by a single fully connected dense layer parameterised by weight matrix $W \in \mathbb{R}^{1280\times 6}$ and bias vector $b \in \mathbb{R}^6$:

$$z(x) = W^\top h(x) + b, \quad z(x) \in \mathbb{R}^6. \tag{6}$$

The discrete target label space encompasses $K = 6$ commercial fruit freshness classes:

$$\begin{aligned} \mathcal{Y} = \{&0 : \text{fresh_apple},\ 1 : \text{fresh_banana},\ 2 : \text{fresh_orange},\\ &3 : \text{rotten_apple},\ 4 : \text{rotten_banana},\ 5 : \text{rotten_orange}\}. \end{aligned} \tag{7}$$

4.2.2. Model Provenance and Immutability Safeguards

The evaluated model (designated Model B V4.1) was trained on $N_{\text{train}} = 9,513$ images from the primary Kaggle archive using AdamW optimization (initial learning rate 10^{-3}, weight decay 10^{-4}, label smoothing $\alpha = 0.05$) under group-aware partition separation. The final model checkpoint is stored as MODEL_B_V4_1_CHECKPOINT.keras and verified by cryptographic checksum:

SHA-256: 4a12f354a4157d2b04ed9606df7112c663fe000d924c5264b5f2fa1b3932b6b4

Throughout all external evaluations across Partitions B, C, D1, and D2, this checkpoint remained strictly frozen ('training=False'). No weights, biases, batch normalization statistics, or hyperparameters were adjusted or fine-tuned.

4.2.3. Input Preprocessing

Input images of arbitrary resolution and aspect ratio are decoded from raw JPEG/PNG formats, resized to $(224, 224, 3)$ using bilinear interpolation, and converted to 32-bit floating-point tensors. Pixel intensities in $[0, 255]$ are normalized according to the standard MobileNetV2 mapping:

$$x_{\text{norm}} = \frac{x}{127.5} - 1.0 \in [-1.0, 1.0]^3. \tag{8}$$

4.3. Post-Hoc OOD Detection Scoring Functions

We evaluate five post-hoc scoring algorithms, covering posterior probabilities, energy modeling, input perturbation, distance-based feature modeling, and extreme value theory, noting that two evaluated methods are specific implementation variants (an ODIN-style perturbation variant and a modified OpenMax-style heuristic). Table 5 summarizes their mathematical definitions, input representations, score orientations, and implementation hyperparameters.

4.3.1. Maximum Softmax Probability (MSP)

The baseline detector proposed by Hendrycks and Gimpel [9] computes the maximum posterior probability across known classes:

$$p_k(x) = \frac{\exp(z_k(x))}{\sum_{j=1}^{K} \exp(z_j(x))}, \quad S_{\text{MSP}}(x) = \max_{k \in \{0,\ldots,K-1\}} p_k(x). \tag{9}$$

Because softmax normalization enforces $\sum_k p_k(x) = 1$, the score $S_{\text{MSP}}(x) \in [1/K, 1]$ directly reflects the classifier's relative confidence. Higher MSP values indicate in-distribution confidence; lower values indicate uncertainty. For AUROC calculations where out-of-distribution instances represent the positive class ($y_{\text{ood}} = 1$), the anomaly ranking score is $-S_{\text{MSP}}(x)$.

4.3.2. Energy-Based OOD Detection

To overcome softmax probability saturation on anomalous inputs, Liu et al. [18] proposed an energy-based scoring function derived from the Helmholtz free energy. In physical systems, the free energy $E(x; f)$ is defined as $E(x; f) = -T \cdot \log \sum_k \exp(z_k(x)/T)$. In this study, to maintain mathematical consistency with the operational Dual-Gate acceptance equation ($S(x) \geq \tau$), we define the in-distribution energy score directly as the negative free energy (the log-sum-exp of logits):

$$S_{\text{Energy}}(x) = T \cdot \log \sum_{k=0}^{K-1} \exp\left(\frac{z_k(x)}{T}\right). \tag{10}$$

Table 5. Post-hoc out-of-distribution detector formulations and verified implementation configurations. All scores are oriented such that higher values indicate in-distribution confidence, ensuring consistency across continuous AUROC ranking and operational decision gating.

Scoring Method	Mathematical Formulation	Input Layer	Score Direction	Verified Implementation Settings
MSP Hendrycks & Gimpel [9]	$S_{\mathrm{MSP}}(x) = \max_k \frac{\exp(z_k(x))}{\sum_j \exp(z_j(x))}$	Logits $z(x) \in \mathbb{R}^6$	Higher $\Longrightarrow$ ID Anomaly: $-S_{\mathrm{MSP}}$	Standard softmax posterior; temperature $T = 1.0$; zero hyperparameter tuning.
Energy Liu et al. [18]	$S_{\mathrm{Energy}}(x) = T \cdot \log \sum_{k=1}^{K} \exp(z_k(x)/T)$	Logits $z(x) \in \mathbb{R}^6$	Higher $\Longrightarrow$ ID Anomaly: $-S_{\mathrm{Energy}}$	Free-energy score; temperature $T = 1.0$; computed via logsumexp. Post-hoc score, not generative density $p(x)$.
ODIN-style Variant Liang et al. [16]	$S_{\mathrm{ODIN}}(x) = \max_k \frac{\exp(z_k(\tilde{x})/T)}{\sum_j \exp(z_j(\tilde{x})/T)}$ $\tilde{x} = \mathrm{clip}(x + \varepsilon\, \mathrm{sign}(\nabla_x \max_k z_k/T))$	Image x and logits $z(x)$	Higher $\Longrightarrow$ ID Anomaly: $-S_{\mathrm{ODIN}}$	ODIN-style perturbation variant; gradient on max temperature-scaled logit; $T = 1000.0$; $\varepsilon = 0.0014$; clipped to $[-1.0, 1.0]$.
Ledoit–Wolf Mahalanobis Lee et al. [17]	$S_{\mathrm{Maha}}(x) = -\min_c d_M(h(x), \mu_c)$ $d_M = \sqrt{(h(x)-\mu_c)^\top \Sigma_{\mathrm{LW}}^{-1} (h(x)-\mu_c)}$	Features $h(x) \in \mathbb{R}^{1280}$	Higher $\Longrightarrow$ ID Anomaly: $d_M(x)$	Class centroids μ_c; shared covariance with Ledoit-Wolf shrinkage Σ_{LW}; negated distance.
Modified OpenMax-style Bendale & Boult [14]	$S_{\mathrm{OpenMax}}(x) = 1.0 - p_{\mathrm{OM}}(y=0)$ $w_{c_r} = 1.0 - \frac{r-1}{\alpha}(1-\omega_r)$	Features $h(x)$ and logits $z(x)$	Higher $\Longrightarrow$ ID Anomaly: $p_{\mathrm{OM}}(0) - 1$	Modified OpenMax-style heuristic; 1280-D global pooled features; Weibull tail $\eta = 20$; $\alpha = 2$; rank weighting gives top-1 class zero logit discount.

In our implementation, the temperature is set to $T = 1.0$, simplifying the formulation to $S_{\mathrm{Energy}}(x) = \mathrm{logsumexp}(z(x))$. Higher energy scores indicate larger unnormalized logit magnitudes and higher in-distribution confidence. We emphasize that $S_{\mathrm{Energy}}(x)$ represents a post-hoc discriminator score, and is not claimed to represent an exact generative data density $p(x)$.

4.3.3. ODIN-Style Perturbation Variant

The ODIN framework proposed by Liang et al. [16] combines temperature scaling with adversarial input perturbation to enlarge score separation between in-distribution and anomalous inputs. In canonical ODIN, the input perturbation is derived from the gradient of the negative log temperature-scaled softmax probability of the predicted class: $-\log S_{\hat{y}}(x; T_{\mathrm{ODIN}})$. In our canonical benchmark implementation (scripts/run_stage4_1_benchmark.py), an ODIN-style perturbation variant is evaluated where the gradient target is the maximum temperature-scaled logit:

1. Input Perturbation: The input image is perturbed along the sign of the gradient of the maximum temperature-scaled logit:

$$x_{\mathrm{perturbed}} = \mathrm{clip}\left(x + \varepsilon_{\mathrm{ODIN}} \cdot \mathrm{sign}\left(\nabla_x \max_k \frac{z_k(x)}{T_{\mathrm{ODIN}}}\right),\ -1.0,\ 1.0\right). \tag{11}$$

Because input imagery is preprocessed via the standard MobileNetV2 normalization mapping $x_{\mathrm{norm}} \in [-1.0, 1.0]^3$, perturbed values are strictly clipped to the valid dynamic range $[-1.0, 1.0]$.

2. Temperature-Scaled Confidence Scoring: The perturbed image $x_{\text{perturbed}}$ is passed through the network, and the score is computed as the maximum temperature-scaled softmax probability:

$$S_{\text{ODIN}}(x) = \max_{k\in\{0,\ldots,K-1\}} \frac{\exp(z_k(x_{\text{perturbed}})/T_{\text{ODIN}})}{\sum_{j=0}^{K-1}\exp(z_j(x_{\text{perturbed}})/T_{\text{ODIN}})}. \tag{12}$$

In our canonical implementation, hyperparameters are locked at $T_{\text{ODIN}} = 1000.0$ and $\varepsilon_{\text{ODIN}} = 0.0014$, verified against scripts/run_stage4_1_benchmark.py. We explicitly document this implementation as an ODIN-style variant optimizing the top logit directly rather than canonical cross-entropy loss. Higher values denote in-distribution confidence.

4.3.4. Ledoit–Wolf Mahalanobis Distance

Following Lee et al. [17], the Mahalanobis detector models penultimate feature representations using class-conditional Gaussian distributions with a tied covariance matrix.

1. Centroid Estimation: Using training features $h(x) \in \mathbb{R}^{1280}$, class mean vectors are computed as:

$$\mu_c = \frac{1}{N_c}\sum_{i:y_i=c} h(x_i), \quad c \in \{0,\ldots,K-1\}. \tag{13}$$

2. Regularized Covariance Estimation: To ensure invertibility and mitigate singularity in high-dimensional feature space ($D = 1280$), the shared covariance matrix is estimated using the Ledoit–Wolf shrinkage estimator on centered training features:

$$\Sigma_{\text{LW}} = (1-\rho)\Sigma_{\text{sample}} + \rho\,\nu I_D, \tag{14}$$

where $\rho \in [0,1]$ is the analytically optimal shrinkage intensity and $\nu = \frac{1}{D}\operatorname{tr}(\Sigma_{\text{sample}})$.

3. Score Formulation: For test input x, the squared Mahalanobis distance to the closest class centroid is computed:

$$d_M^2(x) = \min_{c\in\{0,\ldots,K-1\}} (h(x)-\mu_c)^\top \Sigma_{\text{LW}}^{-1}(h(x)-\mu_c). \tag{15}$$

To maintain directional alignment with other detectors, the score implemented in the benchmark codebase is the negated square-root distance:

$$S_{\text{Maha}}(x) = -\sqrt{\max\left(d_M^2(x), 0\right)}. \tag{16}$$

While the canonical formulation in Lee et al. [17] (reviewed in Section 2.3) defines the score using the negative quadratic distance $-d_M^2(x)$, the square root is a strictly monotonic transformation on $[0,+\infty)$. Consequently, it preserves identical pairwise sample rankings, AUROC, and FPR@95 operating characteristics while scaling distance linearly with feature deviations. Higher values of $S_{\text{Maha}}(x)$ indicate closer

proximity to training class clusters (in-distribution), whereas large negative values indicate anomalous outliers.

4.3.5. Modified OpenMax-Style Heuristic

Bendale and Boult [14] introduced OpenMax to calibrate classification logits using Extreme Value Theory (EVT) and estimate an explicit open-set unknown probability. In this study, the benchmark evaluates a modified OpenMax-style heuristic characterized by two implementation specifics:

1. Feature Embedding Representation: Class Mean Activation Vectors (MAVs) $\mu_c^{\mathrm{MAV}} \in \mathbb{R}^{1280}$ and Euclidean distance modeling are evaluated on the 1280-dimensional global-average-pooled penultimate feature embedding $h(x) \in \mathbb{R}^{1280}$, rather than the multi-dimensional spatial activation tensors or layer-specific activation vectors of the original 2016 architecture.
2. Weibull Tail Fitting: For each class c, Euclidean distances $\|h(x_i) - \mu_c^{\mathrm{MAV}}\|_2$ are computed over correctly classified training samples. A Weibull distribution $\mathbf{Weibull}(\mathrm{shape}_c, \mathrm{scale}_c)$ is fitted to the $\eta = 20$ largest distances (tail size $\eta = 20$, location parameter fixed at 0).
3. Rank-Weighting Formulation and Deviation: At inference, classes are ranked in descending order of logits $z_{(1)}(x) \geq z_{(2)}(x) \geq \cdots \geq z_{(K)}(x)$. For the top $\alpha = 2$ ranked classes (with 1-based rank index $r \in \{1, \ldots, \alpha\}$, corresponding to 0-based index $r - 1$), the Weibull survival probability is evaluated:

$$\omega_r(x) = \exp\left(-\left(\frac{\|h(x) - \mu_{c_r}^{\mathrm{MAV}}\|_2}{\mathrm{scale}_{c_r} + 10^{-8}}\right)^{\mathrm{shape}_{c_r}}\right). \tag{17}$$

The logit adjustment weight implemented in the canonical codebase (scripts/run_stage4_1_benchmark.py) is:

$$w_{c_r} = 1.0 - \frac{r-1}{\alpha}(1.0 - \omega_r(x)), \tag{18}$$

while unranked classes retain $w_k = 1.0$. Crucially, under this formulation, the highest-ranked class ($r = 1$) receives $\frac{1-1}{\alpha} = 0$, yielding $w_{c_1} = 1.0$ regardless of $\omega_1(x)$. Consequently, the top-1 predicted logit undergoes zero logit discounting, and Weibull probability recalibration affects only the second-ranked class ($r = 2$, receiving $w_{c_2} = 1.0 - 0.5(1.0 - \omega_2(x))$). This represents a distinct modified OpenMax-style heuristic compared to canonical OpenMax, where the top-ranked class receives the strongest Weibull tail calibration.
4. Synthetic Unknown Class Softmax: An augmented pseudo-logit for the unknown class (index 0) is constructed as:

$$\hat{v}_0(x) = \sum_{k=0}^{K-1} z_k(x)(1.0 - w_k), \quad \hat{v}_{k+1}(x) = z_k(x) \cdot w_k. \tag{19}$$

Applying softmax across the resulting $K + 1$ logits produces posterior probability vector $p_{\mathrm{OM}} \in \Delta^K$. The in-distribution confidence score is defined as:

$$S_{\mathrm{OpenMax}}(x) = 1.0 - p_{\mathrm{OpenMax}}(y = 0). \tag{20}$$

Higher values indicate that the produce belongs to one of the six known classes; lower values reflect high open-set unknown probability.

4.4. Operational Energy–MSP Dual-Gate Cascade

In agricultural packaging and sorting operations, continuous anomaly scores must be converted into deterministic routing decisions: either accepting the produce item for automated packaging under the classifier's top-1 predicted label, or rejecting the item to an abstention bin for manual inspection.

4.4.1. Motivation and Cascade Formulation

Empirical analysis reveals that standalone detectors exhibit complementary failure modes: Energy scores capture unnormalized activation magnitude and separate coarse semantic out-of-distribution produce, whereas MSP reflects posterior distribution peakedness and discriminates ambiguous in-distribution boundaries. The Dual-Gate cascade integrates both criteria into an operational decision heuristic.

We emphasize that the Dual-Gate cascade is an operational decision rule, not a novel standalone mathematical model or learning algorithm. For any incoming image x, the discrete acceptance decision is defined as:

$$\text{ACCEPT}(x) \iff S_{\text{Energy}}(x) \geq \tau_E \ \wedge \ S_{\text{MSP}}(x) \geq \tau_M. \tag{21}$$

Otherwise:

$$\text{REJECT}(x) \iff S_{\text{Energy}}(x) < \tau_E \ \vee \ S_{\text{MSP}}(x) < \tau_M. \tag{22}$$

When $\text{ACCEPT}(x) = 1$, the model assigns its closed-set predicted class:

$$\hat{y}(x) = \arg\max_{k \in \{0,\ldots,5\}} z_k(x). \tag{23}$$

When $\text{REJECT}(x) = 1$, the system abstains from classification and routes the image record to manual review ($\hat{y}(x) = \text{Ⓡ}$).

4.4.2. Continuous Margin Proxy for Exploratory AUROC

Because the Dual-Gate rule evaluates a Boolean conjunction ($g_E(x) \wedge g_M(x)$), it produces a discrete binary decision ($\{0, 1\}$) rather than a continuous ranking score, and therefore cannot directly generate an ROC curve without a continuous ranking proxy. To permit exploratory comparative ROC analysis in benchmark tables, we compute a continuous linear margin proxy:

$$S_{\text{proxy}}(x) = (S_{\text{Energy}}(x) - \tau_E) + (S_{\text{MSP}}(x) - \tau_M). \tag{24}$$

We emphasize that this continuous linear margin proxy is strictly an exploratory metric. Because raw Energy scores ($S_{\text{Energy}}(x) = \text{logsumexp}(z(x))$) are mathematically unbounded on $(-\infty, +\infty)$ with an observed empirical range of $[-1.753, 5.264]$ across the 13,465 evaluation images, whereas MSP scores are strictly bounded in $[1/6, 1]$ (empirically observed $[0.231, 1.000]$), this unweighted additive combination is inherently scale-dependent and dominated by Energy variations. The proxy does not constitute a canonical continuous score

Table 6. Validation-based operating threshold calibration on primary validation data ($N_{\text{val}} = 227$ raw images). All thresholds were locked a priori using empirical order statistics and held fixed throughout external evaluations. Dual-Gate thresholds are jointly calibrated to account for rejection overlap, achieving matched 95.1542% empirical in-distribution coverage.

Configuration / Detector	Score Symbol	Locked Threshold (τ_{95})	Order-Statistic Index ($s_{(k)}$)	Validation Rejected / Accepted	Empirical ID Coverage	Operational Comparison Rule
MSP (Single Gate)	S_{MSP}	0.700580	12th smallest $s_{(12)}$ (index 11)	11 rejected / 216 accepted	95.1542%	ACCEPT $\iff$ $S_{\text{MSP}} \geq 0.700580$
Energy (Single Gate)	S_{Energy}	1.054579	12th smallest $s_{(12)}$ (index 11)	11 rejected / 216 accepted	95.1542%	ACCEPT $\iff$ $S_{\text{Energy}} \geq 1.054579$
ODIN-style Variant (Single Gate)	S_{ODIN}	0.167164	12th smallest $s_{(12)}$ (index 11)	11 rejected / 216 accepted	95.1542%	ACCEPT $\iff$ $S_{\text{ODIN}} \geq 0.167164$
Ledoit–Wolf Mahalanobis (Single Gate)	S_{Maha}	-65.133230	12th smallest $s_{(12)}$ (index 11)	11 rejected / 216 accepted	95.1542%	ACCEPT $\iff$ $S_{\text{Maha}} \geq -65.133230$
Modified OpenMax-style (Single Gate)	S_{OpenMax}	0.778938	12th smallest $s_{(12)}$ (index 11)	11 rejected / 216 accepted	95.1542%	ACCEPT $\iff$ $S_{\text{OpenMax}} \geq 0.778938$
Dual-Gate Cascade (Energy AND MSP)	τ_E and τ_M	τ_E = 0.918451 τ_M = 0.582610	7th smallest $s_{(7)}$ (index 6) on both scores	$\lvert R_1\rvert = 6, \lvert R_2\rvert = 6$ $\lvert R_1 \cap R_2\rvert = 1 \implies$ 11 rejected / 216 accepted	95.1542%	ACCEPT $\iff$ $S_{\text{Energy}} \geq 0.918451 \wedge$ $S_{\text{MSP}} \geq 0.582610$

of the Dual-Gate. All operational routing and acceptance metrics (UFAR, UFAR-Fresh, KRR, Retention) are evaluated strictly using the exact discrete Boolean conjunction in Eq. (21).

4.5. Validation-Based Operating-Threshold Calibration

To reduce potential circularity and data-snooping bias, all operating thresholds were calibrated strictly on an independent primary validation partition and locked prior to external evaluation.

4.5.1. Calibration Population

The calibration population comprises $N_{\text{val}} = 227$ raw photographic images from the primary Kaggle validation split (FINAL_GROUP_VAL_V4_1.csv, filtered by 'is_raw == 1'). This set contains 35 fresh apples, 49 rotten apples, 33 fresh bananas, 46 rotten bananas, 31 fresh oranges, and 33 rotten oranges. As established in Section 3.7, MD5 cryptographic hashing confirms zero file overlap with the 227-image test set (Partition A), and the 227 validation images are completely excluded from the 13,465-image evaluation census.

4.5.2. Order-Statistic Calibration Protocol

Thresholds were derived using non-parametric empirical order statistics to target approximately 95% in-distribution acceptance coverage (τ_{95}, corresponding to an empirical false rejection / abstention budget of $\leq 5.0\%$). Table 6 details the exact calibration parameters, order-statistic indices, and empirical validation coverage for each configuration.

1. Standalone Single-Gate Thresholds: Let $\{s_{(1)} \leq s_{(2)} \leq \cdots \leq s_{(N_{\text{val}})}\}$ denote the sorted validation scores of a detector. Selecting the 12th smallest order statistic ($s_{(12)}$, 0-indexed position 11) establishes a threshold

that rejects the 11 smallest scores ($S < s_{(12)}$) and accepts the remaining $227 - 11 = 216$ samples:

$$\text{Empirical In-Distribution Coverage} = \frac{216}{227} \approx 95.1542\%. \tag{25}$$

Under this rule, standalone Energy yields $\tau_{\text{Energy}} = 1.054579$ and standalone MSP yields $\tau_{\text{MSP}} = 0.700580$.

2. Joint Dual-Gate Calibration via Inclusion–Exclusion: If the standalone thresholds were applied simultaneously in an AND-rule, rejections would compound: $|R_{\text{union}}| = |R_E \cup R_M| > 11$, reducing in-distribution acceptance far below 95%. To maintain matched 95.1542% empirical coverage, the thresholds for both gates are jointly calibrated using empirical set intersection:

   - Gate 1 (Energy) threshold is relaxed to the 7th smallest order statistic ($s_{(7)}$, index 6): $\tau_E = 0.918451$, rejecting $|R_1| = 6$ samples.
   - Gate 2 (MSP) threshold is relaxed to the 7th smallest order statistic ($s_{(7)}$, index 6): $\tau_M = 0.582610$, rejecting $|R_2| = 6$ samples.
   - Exactly 1 sample falls below both thresholds simultaneously: $|R_1 \cap R_2| = 1$.
   - By the inclusion–exclusion principle, the total number of rejected validation samples is:

$$|R_{\text{union}}| = |R_1| + |R_2| - |R_1 \cap R_2| = 6 + 6 - 1 = 11. \tag{26}$$

   - The joint in-distribution acceptance count is exactly $227 - 11 = 216$, matching the canonical 95.1542% empirical validation coverage.

We emphasize that the observed 95.1542% validation coverage represents an empirical sample proportion, not a guaranteed population-level coverage certificate; with $N_{\text{val}} = 227$, the 95% Clopper–Pearson exact binomial interval spans [91.50%, 97.56%] (exact: [91.4951%, 97.5565%]).

4.6. Evaluation Metrics

To evaluate model performance across multiple operational dimensions, we define three categories of metrics, summarized in Table 7.

4.6.1. Closed-Set Classification Metrics

Evaluated strictly on known-class partitions (Partition A, $N = 227$; Partition B, $N = 5,350$):

1. Top-1 Accuracy: $\text{Acc} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\{\hat{y}_i = y_i\}$.
2. Balanced Accuracy: Arithmetic mean of recall across all $K = 6$ classes, preventing majority-class dominance:

$$\text{Balanced Accuracy} = \frac{1}{K}\sum_{k=0}^{K-1} \frac{\text{TP}_k}{N_k}. \tag{27}$$

Table 7. Authoritative evaluation metrics, mathematical formulations, target directions, and operational evaluation scopes across closed-set classification, continuous out-of-distribution ranking, and operational rejection gating.

Metric Name	Category	Mathematical Formulation	Target Direction	Operational Scope and Meaning
Top-1 Accuracy	Closed-Set Classification	$\mathrm{Acc} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\{\hat{y}_i = y_i\}$	Higher (↑)	Overall proportion of correct predictions across known classes $\mathcal{Y}$.
Balanced Accuracy	Closed-Set Classification	$\mathrm{BalAcc} = \frac{1}{K}\sum_{k=0}^{K-1} \frac{\mathrm{TP}_k}{N_k}$	Higher (↑)	Unweighted macro-averaged recall across all 6 known classes; eliminates class-imbalance bias.
Macro-Averaged F1	Closed-Set Classification	$\text{Macro-F1} = \frac{1}{K}\sum_{k=0}^{K-1} \frac{2P_kR_k}{P_k+R_k}$	Higher (↑)	Harmonic mean of class-wise precision and recall; treats all 6 target classes with equal weight.
AUROC	Continuous OOD Ranking	$\mathrm{AUROC} = \int_0^1 \mathrm{TPR}(\tau)\, d(\mathrm{FPR}(\tau))$	Higher (↑)	Area under ROC curve treating OOD as positive ($y_{\text{ood}} = 1$); threshold-independent ranking.
Retrospective FPR@95	Retrospective Operating Point	$\text{FPR@95} = \frac{\lvert\{x\in\mathcal{D}_{\text{ood}} : S(x) \geq \tau_{95}^{\text{test}}\}\rvert}{\lvert\mathcal{D}_{\text{ood}}\rvert}$	Lower (↓)	False positive rate on negative class (OOD) when τ_{95}^{test} fixes positive class (ID) TPR at 95.0%; retrospective operating-point metric.
OSCR	Open-Set Classification	$\mathrm{OSCR} = \int_0^1 \mathrm{CCR}(\tau)\, d(\mathrm{FPR}(\tau))$	Higher (↑)	Area under Correct Classification Rate vs. False Positive Rate curve.
Operational UFAR	Operational Rejection Gating	$\mathrm{UFAR}(\tau_{95}) = \frac{\lvert\{x\in\mathcal{D}_{\text{ood}} : \text{ACCEPT}(x)\}\rvert}{\lvert\mathcal{D}_{\text{ood}}\rvert}$	Lower (↓)	Proportion of out-of-distribution image records accepted under fixed validation threshold τ_{95}.
Operational UFAR-Fresh	Produce Routing Outcome	$\text{UFAR-F}(\tau_{95}) = \frac{\lvert\{x\in\mathcal{D}_{\text{ood}} : \text{ACCEPT}(x) \wedge \hat{y}(x)\in\text{Fresh}\}\rvert}{\lvert\mathcal{D}_{\text{ood}}\rvert}$	Lower (↓)	Proportion of OOD image records accepted by the evaluated gate and assigned one of the classifier's Fresh-class output labels ($\hat{y} \in \{0, 1, 2\}$); image-level non-target acceptance outcome, not verified biological food safety or grade certification.
Operational KRR (Abstention Rate)	Operational Inspection Cost	$\mathrm{KRR}(\tau_{95}) = \frac{\lvert\{x\in\mathcal{D}_{\text{id}} : \text{REJECT}(x)\}\rvert}{\lvert\mathcal{D}_{\text{id}}\rvert}$	Lower (↓)	Known-class rejection rate: proportion of known-class image records routed to manual inspection.
Known Retention	Operational Automation	$\text{Retention} = 1 - \mathrm{KRR}(\tau_{95})$	Higher (↑)	Proportion of known-class image records retained and classified automatically without human review.

3. Macro-Averaged F1: Unweighted average of class-wise F1 scores:

$$\text{Macro-F1} = \frac{1}{K}\sum_{k=0}^{K-1} \frac{2 \cdot \text{Precision}_k \cdot \text{Recall}_k}{\text{Precision}_k + \text{Recall}_k}. \tag{28}$$

4.6.2. Continuous Out-of-Distribution Discrimination Metrics

Evaluated across in-distribution and out-of-distribution subsets ($\mathcal{D}_{\text{id}} \cup \mathcal{D}_{\text{ood}}$):

1. Area Under the ROC Curve (AUROC): Measures threshold-independent pairwise ranking capability. In our evaluation protocol, out-of-distribution instances represent the positive class ($y_{\text{ood}} = 1$) and in-distribution known instances represent the negative class ($y_{\text{id}} = 0$). With detector score $S(x)$ oriented such that higher values denote in-distribution, the anomaly score is defined as $-S(x)$. The AUROC equals the probability that a randomly drawn OOD sample receives a higher anomaly score than a randomly drawn ID sample:

$$\text{AUROC} = P(-S(x_{\text{ood}}) > -S(x_{\text{id}})) = P(S(x_{\text{id}}) > S(x_{\text{ood}})). \tag{29}$$

2. Retrospective FPR@95: The false positive rate (proportion of OOD images incorrectly accepted as known) when a threshold τ_{95}^{test} is chosen post-hoc on the evaluated test set to achieve at least 95.0% true positive rate (known retention) on $\mathcal{D}_{\text{id}}$:

$$\tau_{95}^{\text{test}} = \sup\left\{\tau : \frac{|\{x \in \mathcal{D}_{\text{id}} : S(x) \geq \tau\}|}{|\mathcal{D}_{\text{id}}|} \geq 0.95\right\}, \quad \text{FPR@95} = \frac{|\{x \in \mathcal{D}_{\text{ood}} : S(x) \geq \tau_{95}^{\text{test}}\}|}{|\mathcal{D}_{\text{ood}}|}. \tag{30}$$

   In this operating-point formulation, in-distribution (ID) produce represents the positive class, and out-of-distribution (OOD) produce represents the negative class. Setting the retrospective threshold τ_{95}^{test} enforces an in-distribution True Positive Rate of at least 95.0% ($\text{TPR}_{\text{id}} \geq 0.95$), and the false positive rate on OOD measures the proportion of negative (OOD) image records incorrectly accepted as positive (ID). This contrasts directly with AUROC, where out-of-distribution instances are designated as the positive target class ($y_{\text{ood}} = 1$) and evaluated across all possible operating thresholds using the anomaly score $-S(x)$.
3. Auxiliary Open-Set Discrimination Metrics: In addition to primary ranking and separation metrics, exploratory discrimination diagnostics were evaluated during execution: the Correct Classification Rate vs. False Positive Rate curve ($\text{OSCR} = \int_0^1 \text{CCR}(\tau)\, d(\text{FPR}(\tau))$) and precision-recall curves (AUPR-IN treating knowns as positive; AUPR-OUT treating OOD as positive), preserved in STAGE4_1_2_SOURCE_STRATIFIED_METRICS.csv.

4.6.3. Operational Decision Metrics under Locked Validation Thresholds

Evaluated strictly using the prospective thresholds (τ_{95}) locked on the Kaggle validation set:

1. Unknown False Acceptance Rate (UFAR): The proportion of out-of-distribution image records accepted as known:

$$\text{UFAR}(\tau_{95}) = \frac{|\{x \in \mathcal{D}_{\text{ood}} : \text{ACCEPT}(x)\}|}{|\mathcal{D}_{\text{ood}}|}. \tag{31}$$

2. Unknown-to-Fresh False Acceptance Rate (UFAR-Fresh): The proportion of out-of-distribution image records accepted and specifically assigned to a model-predicted Fresh class ($\hat{y}(x) \in \{0, 1, 2\}$, corresponding

to fresh apple, fresh banana, or fresh orange):

$$\text{UFAR-Fresh}(\tau_{95}) = \frac{|\{x \in \mathcal{D}_{\text{ood}} : \text{ACCEPT}(x) \ \wedge \ \hat{y}(x) \in \{0,1,2\}\}|}{|\mathcal{D}_{\text{ood}}|}. \tag{32}$$

Crucially, the denominator of UFAR-Fresh is the total number of OOD images ($|\mathcal{D}_{\text{ood}}|$), not the accepted subset, providing an absolute failure rate across the evaluated cohort. We define UFAR-Fresh strictly as the proportion of OOD image records accepted by the evaluated gate and assigned one of the classifier's Fresh-class output labels; it is not claimed to represent verified biological freshness, microbial non-contamination, toxicological clearance, or commercial quality certification.

3. Known-Class Rejection Rate (KRR / Abstention Rate): The proportion of known-class image records rejected by the gate:

$$\text{KRR}(\tau_{95}) = \frac{|\{x \in \mathcal{D}_{\text{id}} : \text{REJECT}(x)\}|}{|\mathcal{D}_{\text{id}}|}. \tag{33}$$

4. Known-Class Retention Rate: The proportion of known-class image records retained for automated sorting: Retention = $1 - \text{KRR}(\tau_{95})$.

4.7. Uncertainty Quantification and Statistical Reporting

All reported confidence intervals and empirical uncertainty summaries adhere to pre-specified, non-parametric protocols:

1. Non-Parametric Bootstrap Protocol: Point estimates are accompanied by 95% bootstrap confidence intervals generated via $B = 1,000$ resamples with replacement. To support repeatability of the bootstrap resampling procedure across runs, pseudo-random number generation was initialized with 'seed=42'. Confidence intervals are constructed using the empirical percentile method:

$$\text{CI}_{95} = \left[\hat{\theta}^*_{(0.025 \cdot B)}, \ \hat{\theta}^*_{(0.975 \cdot B)}\right] = \left[\hat{\theta}^*_{(25)}, \ \hat{\theta}^*_{(975)}\right]. \tag{34}$$

2. Resampling Unit and Stratification Strategy: The resampling unit is strictly an individual image record. For closed-set metrics on Partitions A and B, bootstrap resampling is stratified by the $K = 6$ known classes. For out-of-distribution cohort evaluations (Cohorts A–D), resampling is stratified jointly across the reference ID population and the OOD cohort by species and condition labels.
3. Methodological Conditioning and Statistical Non-Independence: All bootstrap confidence intervals are conditional on image-level exchangeability. As detailed in Section 3.6, photographic records from web archives cannot be assumed to represent statistically independent biological fruit specimens. Crucially, Cohorts A–C share the same underlying FruitVision Known reference population ($N_{\text{id}} = 4,157$). Although bootstrap resampling was performed independently for each cohort evaluation ($B = 1,000$, seed 42), the resulting marginal intervals are not paired confidence intervals for differences between cohorts. Consequently, these cohorts should not be treated as independent experimental replications, and their reported point estimates and confidence intervals share an underlying empirical reference sample.
4. Descriptive Comparison of Closed-Set Performance Differences: The observed 28.56 percentage-point difference in top-1 accuracy between Partition A (96.92% [94.71%, 98.68%]) and Partition B (68.36% [67.20%, 69.40%]) displays non-overlapping 95% bootstrap percentile intervals, reflecting substantial

descriptive separation under the evaluated image-level resampling protocol without asserting formal inferential hypothesis testing.

4.8. Software and Implementation Reproducibility

To ensure complete computational traceability, all experimental code was executed within a controlled software environment:

- Software Stack: Python 3.10.12, TensorFlow 2.15.0, Keras 3.0.5, NumPy 1.26.4, SciPy 1.13.0, scikit-learn 1.4.1.post1, and pandas 2.2.1.
- Image Processing Pipeline: Standardized via tf.io.decode_image, tf.image.resize (bilinear), and tf.keras.applications.mobilenet_v2.preprocess_input.
- OOD Implementation Libraries: SciPy scipy.special.logsumexp for Energy calculation; SciPy scipy.stats.weibull_min for OpenMax Weibull tail fitting; scikit-learn Ledoit–Wolf shrinkage covariance estimation for Mahalanobis scoring; and TensorFlow tf.GradientTape for ODIN input perturbations.
- Canonical Deliverable Ledgers: Execution outputs are preserved in structured files within the repository root:
  - Master image-level prediction and score table: STAGE4_1_IMAGE_LEVEL_PREDICTIONS.csv
  - Frozen validation operating thresholds: STAGE4_0_3_CORRECTED_THRESHOLD_TABLE.csv
  - Source-stratified point estimates: STAGE4_1_2_SOURCE_STRATIFIED_METRICS.csv
  - Authoritative bootstrap confidence intervals ($B = 1,000$): STAGE4_1_2_1_SOURCE_STRATIFIED_CONFIDENCE_INTERVALS.csv
  - Perceptual hash sensitivity evaluation tiers: STAGE4_1_2_1_SOURCE_STRATIFIED_PHASH_SENSITIVITY.csv
  - Final claims and evidence ledger: STAGE4_1_2_1_1_FINAL_CLAIMS_LEDGER.csv
- External Dependencies: Full end-to-end re-execution requires access to the original photographic files distributed by Kaggle, FruitVision [12], and Sultana et al. [11] under their respective publisher terms.

5. Experimental Results and Failure Analysis

This section presents the closed-set performance evaluation across partitions, source-stratified OOD detection AUROC and FPR@95 metrics, operational Dual-Gate rejection rates (UFAR, UFAR-Fresh, KRR), non-parametric bootstrap confidence intervals, and sensitivity analysis across perceptual-hash (pHash) near-duplicate exclusion tiers.

5.1. Sensitivity and Near-Duplicate Robustness Analysis

To evaluate whether near-duplicate captures within external photographic archives artificially inflate or distort out-of-distribution detection metrics, detector performance was benchmarked across the three sensitivity tiers defined in Section 3.6: Primary ($N = 13,465$), Tier 1 ($N = 13,409$), and Tier 2 ($N = 13,382$). Table 8 reports the complete empirical sensitivity evaluation across all five post-hoc scoring functions and the exploratory Dual-Gate linear-margin proxy for all four source-matched cohorts. We explicitly qualify that the Dual-Gate AUROC metric represents the exploratory continuous linear-margin proxy ($S_{\text{proxy}} = (S_E - \tau_E) + (S_M - \tau_M)$), whose unweighted formulation is scale-dependent and dominated by Energy variance ($\sigma \approx 2.15$ vs. $\sigma \approx 0.18$ for MSP), rather than the discrete operational Boolean AND rule itself.

The sensitivity analysis demonstrates two critical empirical findings:

Table 8. Sensitivity analysis of out-of-distribution detection performance (AUROC, %) across perceptual-hash (pHash) near-duplicate exclusion tiers. Results cover all five post-hoc scoring functions and the exploratory Dual-Gate linear margin proxy across Cohorts A–D. Relative detector rankings remain strictly invariant across all three tiers within every evaluation cohort.

Evaluation Cohort	Scoring Method	Primary ($N = 13,465$)	Tier 1 ($N = 13,409$)	Tier 2 ($N = 13,382$)	Max $\lvert\Delta\rvert$ (pp)
Cohort A: Semantic Shift	MSP	72.42	72.38	72.33	0.09
FruitVision B vs D1	Energy	81.57	81.53	81.46	0.11
$N_{\text{id}} : 4,157 \to 4,146 \to 4,140$	ODIN (Variant)	72.30	72.27	72.20	0.10
$N_{\text{ood}} : 2,776 \to 2,762 \to 2,746$	LW-Mahalanobis	81.41	81.34	81.27	0.14
	OpenMax (Mod.)	72.46	72.44	72.38	0.08
	Dual-Gate (Proxy)	80.93	80.89	80.82	0.11
Cohort B: Treatment Shift	MSP	58.77	58.80	58.81	0.04
FruitVision B vs C	Energy	53.13	53.18	53.19	0.06
$N_{\text{id}} : 4,157 \to 4,146 \to 4,140$	ODIN (Variant)	56.73	56.78	56.78	0.05
$N_{\text{ood}} : 1,917 \to 1,906 \to 1,901$	LW-Mahalanobis	47.91	47.96	48.01	0.10
	OpenMax (Mod.)	55.35	55.38	55.40	0.05
	Dual-Gate (Proxy)	53.96	54.01	54.02	0.06
Cohort C: Combined Shift	MSP	76.47	76.48	76.50	0.03
FruitVision B vs D2	Energy	84.72	84.64	84.65	0.08
$N_{\text{id}} : 4,157 \to 4,146 \to 4,140$	ODIN (Variant)	78.11	78.11	78.13	0.02
$N_{\text{ood}} : 1,195 \to 1,175 \to 1,175$	LW-Mahalanobis	80.08	79.88	79.86	0.22
	OpenMax (Mod.)	77.22	77.22	77.22	0.00
	Dual-Gate (Proxy)	84.16	84.09	84.10	0.07
Cohort D: Semantic Shift	MSP	59.77	59.77	59.77	0.00
Sultana B vs D1	Energy	67.52	67.52	67.52	0.00
$N_{\text{id}} : 1,193 \to 1,193 \to 1,193$	ODIN (Variant)	60.02	60.02	60.02	0.00
$N_{\text{ood}} : 2,000 \to 2,000 \to 2,000$	LW-Mahalanobis	68.15	68.15	68.15	0.00
	OpenMax (Mod.)	64.38	64.38	64.38	0.00
	Dual-Gate (Proxy)	66.68	66.68	66.68	0.00

1. High Metric Stability: Across all 24 evaluated detector-cohort configurations, metric fluctuations between the Primary benchmark and Tier 2 remain minor: the maximum observed change in AUROC across all methods is 0.22 percentage points (Ledoit–Wolf Mahalanobis in Cohort C, shifting from 80.08% to 79.86%). For most scoring functions, AUROC shifts are bounded within ±0.11 percentage points. In Cohort D (Sultana et al.), where zero cross-partition or multi-label pHash collisions were detected at $r \leq 8$, metrics remain completely invariant.
2. Strict Detector Ranking Invariance: In all four cohorts, the relative ranking of detection methods is preserved without a single rank inversion across the three evaluation tiers:

   - Cohort A (FruitVision Semantic Shift): Energy (81.57% → 81.46%) > Ledoit–Wolf Mahalanobis (81.41% → 81.27%) > Dual-Gate Proxy (80.93% → 80.82%) > Modified OpenMax (72.46% → 72.38%) > MSP (72.42% → 72.33%) > ODIN Variant (72.30% → 72.20%).
   - Cohort B (FruitVision Treatment Shift): MSP (58.77% → 58.81%) > ODIN Variant (56.73% → 56.78%) > Modified OpenMax (55.35% → 55.40%) > Dual-Gate Proxy (53.96% → 54.02%) > Energy (53.13% → 53.19%) > Ledoit–Wolf Mahalanobis (47.91% → 48.01%).
   - Cohort C (FruitVision Combined Shift): Energy (84.72% → 84.65%) > Dual-Gate Proxy (84.16% → 84.10%) > Ledoit–Wolf Mahalanobis (80.08% → 79.86%) > ODIN Variant (78.11% → 78.13%) > Modified OpenMax (77.22% → 77.22%) > MSP (76.47% → 76.50%).
   - Cohort D (Sultana Semantic Shift): Ledoit–Wolf Mahalanobis (68.15%) > Energy (67.52%) > Dual-Gate Proxy (66.68%) > Modified OpenMax (64.38%) > ODIN Variant (60.02%) > MSP (59.77%) (invariant across all three tiers).

Table 9. Closed-set classification performance across laboratory and external photographic domains. The frozen MobileNetV2 classifier experiences a descriptive 28.56 percentage-point degradation in top-1 accuracy when evaluated on external imagery. Brackets indicate 95% non-parametric bootstrap percentile intervals ($B = 1,000$, seed 42) conditional on image-level resampling.

Evaluation Partition	Upstream Source Description	N	Top-1 Accuracy (%)	Macro-Averaged F1 (%)	Balanced Accuracy (%)
Partition A (ID Test)	Kaggle Primary Benchmark (Lab)	227	96.92 [94.71, 98.68]	96.87 [94.65, 99.05]	97.09 [94.80, 98.98]
Partition B (Pooled External)	FruitVision ($N = 4,157$) + Sultana ($N = 1,193$)	5,350	68.36 [67.20, 69.40]	67.28 [66.21, 68.32]	68.66 [67.55, 69.69]
FruitVision B Subset	FruitVision Known subset (mixed capture settings)	4,157	69.14 [68.13, 70.17]	66.75 [65.62, 68.01]	69.56 [68.52, 70.58]
Sultana B Subset	Sultana Known subset (retail and field)	1,193	65.63 [63.03, 68.15]	65.77 [63.32, 68.13]	65.55 [63.11, 67.99]

Evidence Bounds on Sensitivity Conclusions: These results establish that the benchmark performance estimates and comparative method orderings are robust with respect to the tested pHash collision exclusions and evaluated image records. However, we qualify this conclusion: perceptual hashing at low-frequency DCT resolution captures detectable visual near-duplicates, but cannot prove the complete absence of subtle burst captures, viewpoint variations, or unmodelled specimen-level correlations in the underlying repositories. Complete point estimates and operational metrics across all tiers are preserved in canonical register:
STAGE4_1_2_1_SOURCE_STRATIFIED_PHASH_SENSITIVITY.csv

5.2. Closed-Set Classification Performance Across External Domains

Closed-set classification measures the predictive fidelity of the frozen MobileNetV2 backbone across the six known classes ($K = 6$: Fresh Apple, Rotten Apple, Fresh Banana, Rotten Banana, Fresh Orange, Rotten Orange) under primary in-distribution benchmark conditions (Partition A, $N = 227$) versus external photographic repository conditions (Partition B, $N = 5,350$). Table 9 reports overall top-1 accuracy, macro-averaged F1, and balanced accuracy across partitions alongside their 95% bootstrap confidence intervals.

Under laboratory capture conditions (Partition A), the model achieves 96.92% [94.71%, 98.68%] top-1 accuracy (220/227 correct), 96.87% [94.65%, 99.05%] macro-F1, and 97.09% [94.80%, 98.98%] balanced accuracy, demonstrating strong baseline feature extraction under uniform illumination and neutral backgrounds. However, when evaluated on external repository images (Partition B), top-1 accuracy drops by 28.56 percentage points to 68.36% [67.20%, 69.40%] (3,657/5,350 correct), with macro-F1 falling to 67.28% [66.21%, 68.32%] and balanced accuracy to 68.66% [67.55%, 69.69%]. The 95% bootstrap percentile intervals between Partition A and Partition B exhibit complete non-overlap, reflecting substantial descriptive separation under the evaluated image-level resampling protocol without asserting formal inferential hypothesis testing. Performance remains similarly constrained across both external repositories: 69.14% [68.13%, 70.17%] on FruitVision B ($N = 4,157$) and 65.63% [63.03%, 68.15%] on Sultana B ($N = 1,193$).

To diagnose the drivers of this external degradation, Table 10 breaks down classification performance across individual classes and species. Inspection reveals that domain degradation is highly asymmetric:

1. Catastrophic Fresh Apple Collapse: Fresh Apple accuracy collapses to 32.12% (310/965 correct). The classifier systematically misidentifies 40.73% of fresh apples as Rotten Apple (393 images) and 26.42% as Rotten Orange (255 images). Across the entire Apple species cohort ($N = 1,795$), overall accuracy is only 34.54% (620/1,795).
2. Source-Level Diagnostic Patterns: Within the FruitVision B Fresh Apple subset ($N = 765$), exactly 174 images are correctly classified as Fresh Apple, while 345 are predicted as Rotten Apple, 242 as Rotten Orange, 2 as Fresh Orange, and 2 as Rotten Banana. Within the Sultana B Fresh Orange subset ($N = 193$), 98 images are correct Fresh Orange, while 9 are predicted Fresh Apple, 15 Rotten Apple, and 71 Rotten Orange.

Table 10. Class-wise classification performance and primary misclassification modes on external repository imagery (Partition B, $N = 5,350$). The lowest class-wise accuracy is observed for Fresh Apple (32.12%), which the model systematically confuses with Rotten Apples and Rotten Oranges.

Class / Species Group	Ground Truth (N)	Correct (n)	Accuracy (%)	Primary Empirical Misclassification Modes
Fresh Apple	965	310	32.12	Rotten Apple (393 / 40.73%), Rotten Orange (255 / 26.42%)
Rotten Apple	830	310	37.35	Rotten Orange (361 / 43.49%), Fresh Apple (157 / 18.92%)
Fresh Banana	948	736	77.64	Rotten Banana (207 / 21.84%)
Rotten Banana	831	826	99.40	Fresh Banana (3 / 0.36%)
Fresh Orange	946	828	87.53	Rotten Orange (93 / 9.83%), Fresh Apple (9 / 0.95%)
Rotten Orange	830	647	77.95	Fresh Orange (106 / 12.77%), Rotten Apple (70 / 8.43%)
Species: Apple	1,795	620	34.54	Pervasive decay-state confusion and cross-species misclassification into orange
Species: Banana	1,779	1,562	87.80	High species fidelity; asymmetric bias toward decay over fresh
Species: Orange	1,776	1,475	83.05	Robust species recognition; moderate decay underestimation
Total Partition B	5,350	3,657	68.36	Overall external domain degradation: −28.56 percentage points

3. High Banana and Orange Fidelity: In contrast, Banana classification remains resilient at 87.80% overall accuracy (1,562/1,779), with Rotten Banana achieving 99.40% accuracy (826/831). Orange classification achieves 83.05% overall accuracy (1,475/1,776), though Rotten Orange exhibits modest confusion with Fresh Orange (12.77%).
4. Observational Russeting Hypothesis: Visual inspection of the misclassified FruitVision fresh apple imagery reveals prominent natural surface striations, lenticel spots, and specular reflections characteristic of commercial varietal russeting. We hypothesize that the frozen MobileNetV2 backbone, trained predominantly on smooth, uniform laboratory specimens, correlates surface texture irregularities with fungal decay lesions. We explicitly present this russeting correlation as an observational diagnostic hypothesis rather than a causally verified physiological mechanism.

5.3. Source-Stratified Out-of-Distribution Discrimination

To evaluate out-of-distribution discrimination across structured distribution shifts, detection performance was benchmarked across the four source-matched cohorts defined in Section 3.5. Source matching reduces repository-level provenance differences, while within-repository acquisition variation remains possible. Table 11 presents continuous discrimination metrics (AUROC, retrospective FPR@95, AUPR-IN, AUPR-OUT, and OSCR) across all five post-hoc scoring algorithms and the Energy–MSP linear-margin proxy (exploratory).

The empirical results across the four source-matched cohorts illuminate sharp performance disparities depending on the nature of the distribution shift:

1. Comparative Performance of Energy and Feature Scoring on Semantic Shifts: In Cohort A (FruitVision Semantic Shift), Energy scoring achieves the highest discrimination (AUROC = 81.57% [80.81%, 82.38%], FPR@95 = 68.84%), closely followed by Ledoit–Wolf Mahalanobis (AUROC = 81.41%, FPR@95 = 79.00%) and the Energy–MSP linear-margin proxy (exploratory) (AUROC = 80.93% [80.13%, 81.72%], FPR@95 = 69.27%). Standard MSP (72.42%), the ODIN-style variant (72.30%), and modified OpenMax (72.46%) lag by approximately 9 percentage points in AUROC, with retrospective false positive rates exceeding 82%. In Cohort C (Combined Shift), the observed AUROC for Energy was higher (84.72% [83.85%, 85.58%]) than in Cohort A (81.57%) under the evaluated cohort definitions. However, because Cohort C evaluates a distinct subset of unseen species bearing dataset-

Table 11. Source-stratified out-of-distribution discrimination benchmark across Cohorts A–D. AUROC and retrospective FPR@95 are primary discrimination metrics; AUPR-IN, AUPR-OUT, and OSCR are auxiliary diagnostics. Retrospective FPR@95 measures the proportion of OOD produce incorrectly accepted when thresholding ID knowns at $\mathrm{TPR}_{\mathrm{id}} \geq 95.0\%$. Brackets report 95% bootstrap percentile intervals ($B = 1,000$, seed 42) for Energy and the Energy–MSP linear-margin proxy (exploratory).

Evaluation Cohort	Detection Algorithm	AUROC (%)	FPR@95 (%)	AUPR-IN (%)	AUPR-OUT (%)	OSCR (%)
Cohort A: Semantic FruitVision B vs D1 ID: Apple, Banana, Orange OOD: Grape, Mango $N_{\mathrm{id}} = 4,157, N_{\mathrm{ood}} = 2,776$	MSP (Single Gate)	72.42	82.74	82.04	59.91	55.92
	Energy (Single Gate)	81.57 [80.81, 82.38]	68.84	88.36	72.19	61.28
	ODIN-style Variant (Single Gate)	72.30	79.00	87.65	68.83	54.35
	Ledoit–Wolf Mahalanobis (Single Gate)	81.41	91.28	83.85	55.91	57.26
	Modified OpenMax-style (Single Gate)	72.46	69.27	87.98	71.54	61.11
	Energy–MSP Proxy (Exploratory)	80.93 [80.13, 81.72]				
Cohort B: Condition FruitVision B vs C ID: Natural Knowns OOD: Treated Knowns $N_{\mathrm{id}} = 4,157, N_{\mathrm{ood}} = 1,917$	MSP (Single Gate)	58.77	92.85	78.59	36.73	47.62
	Energy (Single Gate)	53.13 [52.16, 54.09]	94.42	71.25	33.41	43.45
	ODIN-style Variant (Single Gate)	56.73	98.07	67.53	29.60	32.67
	Ledoit–Wolf Mahalanobis (Single Gate)	47.91	95.20	75.43	33.60	45.77
	Modified OpenMax-style (Single Gate)	55.35	94.16	72.03	33.99	44.23
	Energy–MSP Proxy (Exploratory)	53.96 [52.96, 54.91]				
Cohort C: Combined FruitVision B vs D2 ID: Natural Knowns OOD: Treated Unknowns $N_{\mathrm{id}} = 4,157, N_{\mathrm{ood}} = 1,195$	MSP (Single Gate)	76.47	76.40	92.18	47.85	58.01
	Energy (Single Gate)	84.72 [83.85, 85.58]	64.94	95.41	56.71	62.92
	ODIN-style Variant (Single Gate)	78.11	83.85	93.17	43.32	59.69
	Ledoit–Wolf Mahalanobis (Single Gate)	80.08	78.49	93.82	47.10	53.42
	Modified OpenMax-style (Single Gate)	77.22	88.12	93.25	39.71	59.62
	Energy–MSP Proxy (Exploratory)	84.16 [83.28, 85.05]	63.18	95.18	57.63	62.74
Cohort D: Semantic Sultana B vs D1 ID: Apple, Banana, Orange OOD: Guava, Jujube, etc. $N_{\mathrm{id}} = 1,193, N_{\mathrm{ood}} = 2,000$	MSP (Single Gate)	59.77	94.30	53.25	67.96	43.11
	Energy (Single Gate)	67.52 [65.67, 69.25]	83.50	57.10	76.35	46.62
	ODIN-style Variant (Single Gate)	60.02	94.25	52.76	67.88	42.62
	Ledoit–Wolf Mahalanobis (Single Gate)	68.15	81.20	59.17	77.07	42.01
	Modified OpenMax-style (Single Gate)	64.38	94.75	57.23	70.56	45.59
	Energy–MSP Proxy (Exploratory)	66.68 [64.98, 68.38]	84.60	56.54	75.47	46.54

reported treatment labels (Partition D2) rather than an isolated treatment intervention on Cohort A, this comparison reflects the empirical cohort ordering and cannot isolate a causal treatment effect.

2. Weak Discrimination under Treatment-Labelled Condition Shift (Cohort B): Under Cohort B, where identical known species (apple, banana, orange) are evaluated under dataset-reported treatment labels (formalin preservative immersion condition), all evaluated post-hoc scoring algorithms exhibit weak or near-chance discrimination. Energy AUROC is 53.13% [52.16%, 54.09%], the Energy–MSP linear-margin proxy (exploratory) is 53.96% [52.96%, 54.91%], modified OpenMax is 55.35%, the ODIN variant is 56.73%, and MSP is 58.77%. Ledoit–Wolf Mahalanobis falls below chance (AUROC = 47.91%, FPR@95 = 98.07%). Retrospective false positive rates exceed 91% across all methods, indicating that at a 95% known retention operating point, virtually all images bearing dataset-reported treatment labels are accepted into the known class distribution.
3. Domain-Dependent Morphological Confounding (Cohort D): In Cohort D (Sultana Semantic Shift), discrimination is notably lower than in FruitVision Semantic Shift (Cohort A): Ledoit–Wolf Mahalanobis achieves 68.15% AUROC, Energy achieves 67.52% [65.67%, 69.25%], and MSP achieves only 59.77%. Retrospective FPR@95 ranges from 81.20% to 94.75%. As an untested observational hypothesis, this

Table 12. Operational decision performance under prospective validation-calibrated operating thresholds (τ_{95}). Standalone Energy uses $\tau_E = 1.054579$; Dual-Gate Cascade evaluates $S_{\text{Energy}} \geq 0.918451 \wedge S_{\text{MSP}} \geq 0.582610$. Brackets indicate 95% bootstrap percentile intervals ($B = 1,000$, seed 42). UFAR-Fresh measures false acceptance as the proportion of OOD image records accepted by the evaluated gate and assigned one of the classifier's Fresh-class output labels ($\hat{y} \in \{0, 1, 2\}$), rather than commercial quality certification; KRR measures abstention on known-class image records.

Evaluation Cohort	Operational Configuration	Operational UFAR (%)	Operational UFAR-Fresh (%)	Operational KRR (%)	Retained Knowns (%)
Cohort A: Semantic FruitVision B vs D1 $N_{\text{id}} = 4,157, N_{\text{ood}} = 2,776$	Standalone Energy ($\tau_E = 1.054579$)	13.18 [12.07, 14.34] (366)	7.64 [6.70, 8.65] (212)	41.42 [40.34, 42.58] (1,722)	58.58 (2,435)
	Dual-Gate Cascade ($\tau_E \wedge \tau_M$)	16.07 [14.77, 17.40] (446)	10.48 [9.37, 11.60] (291)	41.98 [40.75, 43.06] (1,745)	58.02 (2,412)
Cohort B: Condition FruitVision B vs C $N_{\text{id}} = 4,157, N_{\text{ood}} = 1,917$	Standalone Energy ($\tau_E = 1.054579$)	53.94 [52.16, 55.66] (1,034)	44.03 [42.41, 45.59] (844)	41.42 [40.34, 42.58] (1,722)	58.58 (2,435)
	Dual-Gate Cascade ($\tau_E \wedge \tau_M$)	52.63 [50.81, 54.25] (1,009)	42.83 [41.26, 44.34] (821)	41.98 [40.80, 43.23] (1,745)	58.02 (2,412)
Cohort C: Combined FruitVision B vs D2 $N_{\text{id}} = 4,157, N_{\text{ood}} = 1,195$	Standalone Energy ($\tau_E = 1.054579$)	6.86 [5.52, 8.28] (82)	4.94 [3.85, 6.19] (59)	41.42 [40.34, 42.63] (1,722)	58.58 (2,435)
	Dual-Gate Cascade ($\tau_E \wedge \tau_M$)	9.54 [7.86, 11.30] (114)	7.28 [5.94, 8.70] (87)	41.98 [40.89, 43.20] (1,745)	58.02 (2,412)
Cohort D: Semantic Sultana B vs D1 $N_{\text{id}} = 1,193, N_{\text{ood}} = 2,000$	Standalone Energy ($\tau_E = 1.054579$)	34.15 [32.35, 35.90] (683)	21.45 [19.85, 23.10] (429)	41.74 [39.31, 44.26] (498)	58.26 (695)
	Dual-Gate Cascade ($\tau_E \wedge \tau_M$)	34.90 [33.00, 36.95] (698)	22.65 [20.95, 24.30] (453)	44.01 [41.41, 46.61] (525)	55.99 (668)
Partition B Pooled Pooled External Knowns $N_{\text{id}} = 5,350$	Standalone Energy ($\tau_E = 1.054579$)	— —	— —	41.50 [40.21, 42.69] (2,220)	58.50 (3,130) 57.57 [56.24, 58.90] (3,080)
	Dual-Gate Cascade ($\tau_E \wedge \tau_M$)			42.43 [41.10, 43.76] (2,270)	

lower discrimination may relate to visual commonalities among spherical green fruit classes (such as guava and jujube in Sultana D1 and green apples or citrus in the known set); however, this visual resemblance was not formally quantified or causally tested.

4. Exploratory Energy–MSP Proxy Properties: The Energy–MSP linear-margin proxy (exploratory) achieves AUROC values (80.93% in A, 53.96% in B, 84.16% in C, 66.68% in D) that mirror Energy scoring within 0.84 percentage points across all cohorts. As established in Section 4.4.2, this alignment occurs because unweighted linear combination of unbounded Energy with bounded MSP is mathematically dominated by Energy variance ($\sigma \approx 2.15$ vs. $\sigma \approx 0.18$), yielding a Spearman rank correlation $\rho > 0.992$ between the proxy and raw Energy scores across all 13,465 evaluated image records.

5.4. Operational Rejection and Image-Level Decision Performance

While AUROC evaluates threshold-independent ranking and retrospective FPR@95 evaluates post-hoc operating points on test data, prospective industrial grading systems must operate under fixed thresholds calibrated a priori. Table 12 reports offline operational metrics of standalone Energy and the joint Dual-Gate cascade under locked prospective thresholds calibrated on primary validation data (τ_{95}, targeting 95.15% validation coverage); these metrics evaluate computational image-level acceptance and rejection, not a physical sorting deployment.

Analysis of operational screening reveals notable operational trade-offs and rejection/acceptance outcomes:

1. Severe In-Distribution Abstention Penalty: Operating thresholds calibrated to accept 95.15% of primary in-distribution validation images suffer substantial over-rejection when transferred to external repository images. Standalone Energy exhibits an operational Known-Class Rejection Rate (KRR) of 41.42% on FruitVision B and 41.74% on Sultana B (41.50% pooled). Dual-Gate exhibits a KRR of 41.98% [40.75%, 43.06%] on FruitVision B and 44.01% [41.41%, 46.61%] on Sultana B, pooling at 42.43% [41.10%, 43.76%] (2,270/5,350 known images rejected). Note that because stratified bootstrap resampling was conducted independently per cohort evaluation ($B = 1,000$), resampled draws of the shared in-distribution reference subset (FruitVision B, $N_{\text{id}} = 4,157$) varied across runs, yielding identical point estimates (41.98% for Dual-Gate, 41.42% for standalone Energy) alongside minor variations in empirical percentile bounds across Cohorts A, B, and C. Consequently, retaining automated sorting decisions on only 57.57% of known-class image records presents substantial operational abstention overhead.
2. Operational False Acceptance Rates: On FruitVision Semantic Shift (Cohort A), standalone Energy restricts UFAR to 13.18% [12.07%, 14.34%] (366/2,776 accepted), while Dual-Gate allows 16.07% [14.77%, 17.40%] (446/2,776 accepted). In Cohort C (Combined Shift), UFAR is further suppressed to 6.86% (Energy) and 9.54% (Dual-Gate). However, in Cohort D (Sultana Semantic Shift), operational false acceptance rises steeply to 34.15% (Energy) and 34.90% [33.00%, 36.95%] (Dual-Gate), establishing weaker discrimination and higher operational false acceptance in Cohort D than in Cohort A. While visual commonalities among unseen species (such as guava and jujube in Sultana D1) and known classes represent an untested observational hypothesis, morphological geometry, color distributions, and feature-space overlap were not directly measured, and visual similarity cannot be inferred as the causal driver of this difference.
3. Dataset-Reported Treatment Routing Failure: Under locked validation thresholds, 821 out of 1,917 treatment-labelled image records (42.83% [41.26%, 44.34%]) in FruitVision C are accepted by the Dual-Gate and assigned one of the classifier's Fresh-class output labels ($\hat{y} \in \{0, 1, 2\}$). Standalone Energy exhibits an even higher UFAR-Fresh of 44.03% (844/1,917). Under the evaluated operating thresholds, a substantial fraction of treatment-labelled image records were accepted and assigned one of the classifier's Fresh-class output labels. This outcome does not establish that their confidence-score distributions match those of untreated fresh fruit.
4. Food-Safety Scope Disclaimer: We emphasize that UFAR-Fresh measures strictly the proportion of OOD image records accepted by the evaluated gate and assigned one of the classifier's Fresh-class output labels; it is not claimed to represent verified biological freshness, commercial fruit-grade certification, microbial non-contamination, or toxicological clearance.

5.5. Uncertainty Quantification and Consolidated Benchmark Findings

Synthesizing the empirical evidence from Sections 5.1 through 5.4, we consolidate the benchmark findings with respect to the study's three core research questions:

1. Addressing RQ1 (External Domain Robustness and Spoilage Misclassification): The frozen MobileNetV2 classifier suffers a marked 28.56 percentage-point degradation in top-1 accuracy (from 96.92% to 68.36%) when evaluated on external repository images. The primary failure mode is asymmetric collapse on Fresh Apples (32.12% accuracy; 310/965 correct), where misclassifications frequently fall into decay classes (393 predicted Rotten Apple, 255 Rotten Orange out of 965). Qualitative visual inspection indicates natural surface striations, lenticel spots, and specular reflections consistent with varietal russeting; we present this texture correlation strictly as an untested observational hypothesis rather than a demonstrated causal mechanism.
2. Addressing RQ2 (Comparative Efficacy under Species vs. Condition Shifts): Under source-stratified semantic shift, Energy-based scoring (AUROC = 81.57% on FruitVision, 67.52% on Sultana) and Ledoit–

Wolf Mahalanobis distance (81.41% and 68.15%) substantially outperform standard MSP (72.42% and 59.77%). However, under dataset-reported treatment-labelled condition shift (Cohort B), all evaluated scoring algorithms exhibit weak or near-chance discrimination (Energy AUROC = 53.13%, Mahalanobis = 47.91%, MSP = 58.77%). Tested RGB post-hoc scoring functions did not reliably separate images recorded under the dataset-reported treatment protocol from natural fruit.

3. Addressing RQ3 (Operational Dual-Gate Feasibility and Trade-offs): Validation-calibrated decision gating achieves partial unknown screening on semantic shifts (UFAR = 16.07% in A, 9.54% in C), but imposes a substantial operational abstention burden: 42.43% of known-class image records are rejected / abstained across external repository conditions (KRR = 41.98% on FruitVision, 44.01% on Sultana). Crucially, 42.83% of dataset-reported treatment-labelled image records (821/1,917) are accepted by the Dual-Gate and assigned one of the classifier's Fresh-class output labels, demonstrating that the evaluated visual OOD gates did not provide reliable screening on this treatment cohort. Because physical conveyance, staffing overhead, and financial expenditure were not measured, evaluating real-world operational trade-offs requires dedicated field validation.
4. Methodological Conditioning and Statistical Non-Independence: All reported confidence intervals are conditional on image-level resampling exchangeability ($B = 1,000$). Although Cohorts A–C share the same underlying FruitVision known-class reference records ($N_{\text{id}} = 4,157$), bootstrap resampling was performed independently for each cohort evaluation. Therefore, the marginal confidence intervals should not be interpreted as paired intervals for differences between cohorts. These cohorts are not independent experimental replications.

6. Discussion

The empirical findings from our source-stratified benchmark carry important implications for agricultural computer vision, out-of-distribution detection, and automated post-harvest sorting workflows. In this section, we analyze the structural drivers of external domain collapse, the weak discrimination observed on dataset-reported treatment labels, the potential operational trade-offs of conservative rejection gating, and prospective engineering pathways for multi-sensor quality assessment.

6.1. Domain Collapse and the Brittle Nature of Laboratory Vision Models

A central finding of this study is the marked performance collapse of the frozen MobileNetV2 classifier when transitioned from curated laboratory test data (Partition A, 96.92% accuracy) to external repository environments (Partition B, 68.36% accuracy, representing a 28.56 percentage-point degradation). Both external evaluation subsets independently demonstrate this ceiling: 69.14% on FruitVision B and 65.63% on Sultana B.

Crucially, this degradation is not distributed uniformly across classes. As documented in Section 5.2, the classifier exhibits an asymmetric vulnerability on Fresh Apples, collapsing to 32.12% top-1 accuracy (310/965 correct across pooled Partition B). In pooled Partition B, the model misclassifies 40.73% of fresh apples as Rotten Apple (393 images) and 26.42% as Rotten Orange (255 images). Within the FruitVision B subset alone, 345 fresh apples are predicted as Rotten Apple and 242 as Rotten Orange out of 765 images. Qualitative visual inspection indicates that external photographic records frequently exhibit natural surface striations, lenticel spots, and varietal russeting. In contrast, the laboratory-curated Kaggle training distribution consists predominantly of smooth, uniformly illuminated, blemish-free fruit against neutral planar backgrounds. While this visual comparison suggests that surface texture variations may contribute to false-decay classifications, we distinguish this qualitative observation from a formally tested causal mechanism.

This failure mode highlights a pervasive vulnerability in agricultural computer vision literature: benchmark evaluations conducted solely within closed, laboratory-collected datasets create an impression of near-perfect accuracy (often exceeding 95% to 99%) that degrades substantially when models confront external real-world distribution shifts.

6.2. Weak Discrimination on Dataset-Reported Treatment Labels

Under Cohort B, where identical known species (apple, banana, orange) are evaluated under dataset-reported treatment labels (formalin preservative immersion condition), all five evaluated post-hoc scoring algorithms exhibit weak or near-chance discrimination: Energy AUROC is 53.13% [52.16%, 54.09%], the Energy–MSP linear-margin proxy (exploratory) is 53.96% [52.96%, 54.91%], MSP is 58.77%, and Ledoit–Wolf Mahalanobis distance falls to 47.91%, with retrospective false positive rates exceeding 91% to 98%.

Under the locked prospective operating thresholds (τ_{95}), 821 out of 1,917 dataset-reported treatment-labelled images (42.83% [41.26%, 44.34%]) are accepted by the Dual-Gate and assigned one of the classifier's Fresh-class output labels. Standalone Energy exhibits an even higher rate of acceptance and assignment to Fresh-class output labels of 44.03% (844/1,917).

We emphasize that this study evaluated digital photographic image records bearing repository-assigned labels; we did not perform chemical or toxicological assays to independently verify chemical immersion, confirm compound presence, or quantify tissue residue. Furthermore, our image-based evaluation cannot determine whether biochemical differences manifested as detectable optical features in the evaluated photographs. The empirical finding establishes strictly that within this specific dataset-reported cohort, the evaluated post-hoc vision scoring algorithms did not reliably separate images recorded under the reported treatment protocol from natural known-class fruit. While external RGB photography records visible surface reflectance, this evaluation does not support universal assertions regarding RGB imaging capabilities; rather, on this evaluated dataset-reported cohort, the tested post-hoc visual gates did not provide an effective safeguard against produce carrying treatment labels.

6.3. Operational Trade-offs: The Potential Economic Overhead of Abstention

Our findings illustrate the divergence between continuous discrimination metrics (AUROC) and the practical demands of operational decision gating. While threshold-independent AUROC indicates partial semantic separation on FruitVision (81.57% for Energy), implementing a prospective operational gate under locked validation thresholds (τ_{95}) reveals a substantial abstention overhead.

Operating thresholds calibrated to accept 95.15% of primary in-distribution validation images exhibit an offline operational Known-Class Rejection Rate (KRR) of 41.98% on FruitVision B and 44.01% on Sultana B (42.43% pooled across Partition B). In practical sorting environments, rejecting over four out of ten known-class items represents a high abstention rate with potential operational consequences: routing 42.43% of known-class image records to manual secondary inspection could potentially increase staffing requirements and reduce line throughput, though physical throughput and financial costs were not directly measured in this offline study.

Furthermore, on Cohort D (Sultana Semantic Shift), offline false acceptance rises to 34.90% for Dual-Gate (with 22.65% accepted and assigned one of the classifier's Fresh-class output labels) alongside a 44.01% KRR. As an untested observational hypothesis, visual commonalities among spherical green fruit classes (such as guava and jujube in Sultana D1 and green apples or citrus in the known set) may relate to this difficulty; however, this morphological resemblance was not causally verified, and no dedicated latent embedding-overlap analysis was performed. These results show that, under the locked thresholds, the evaluated score-based gate did not reliably reject all unseen-species records in Cohort D. The underlying score-distribution and feature-space overlap were not independently quantified.

6.4. Architectural Comparison: Post-Hoc Scoring Paradigms

Comparing the post-hoc detection mechanisms reveals clear structural distinctions:

1. Energy vs. Posterior Probability (MSP): Across both semantic shift cohorts (Cohorts A and D), Energy-based scoring outperforms standard MSP, improving AUROC from 72.42% to 81.57% on FruitVision (+9.15 pp) and from 59.77% to 67.52% on Sultana (+7.75 pp). This is consistent with findings that computing the log-sum-exp over unnormalized logits avoids the probability saturation and artificially overconfident posteriors induced by softmax normalization on out-of-distribution inputs [18].

2. Penultimate Feature Modeling (Mahalanobis): Ledoit–Wolf Mahalanobis distance achieves competitive discrimination on semantic shifts (81.41% in A, 68.15% in D). By modeling class-conditional Gaussian clusters in 1280-dimensional feature space, it captures spatial outlier geometry. However, on treatment shift (Cohort B), Mahalanobis distance drops to 47.91% AUROC. This outcome is consistent with weak separation under the evaluated regularized Mahalanobis scoring function. We emphasize that feature-space geometry and cluster overlap were not independently quantified via embedding distances or density estimation; hence, AUROC performance alone cannot be used to infer the exact topological configuration of representations in latent space.
3. Operational Dual-Gate Behavior: The joint Boolean conjunction rule ($S_{\text{Energy}} \geq \tau_E \wedge S_{\text{MSP}} \geq \tau_M$) enforces multi-criteria filtering, suppressing offline UFAR to 16.07% in Cohort A and 9.54% in Cohort C. However, because both constituent gates operate on downstream functions of the same MobileNetV2 logit activations, the cascade remains fundamentally bounded by the backbone's visual representation.
4. Proxy Properties: Continuous ranking analysis for the Dual-Gate evaluates the Energy–MSP linear-margin proxy (exploratory). As established in Section 4.4.2, this proxy correlates strongly with raw Energy scores (Spearman rank correlation $\rho > 0.992$ across all 13,465 evaluated image records), reflecting its mathematical domination by Energy variance ($\sigma \approx 2.15$ vs. $\sigma \approx 0.18$).

6.5. Prospective Pathways for Multi-Modal Sensing in Post-Harvest Quality Assessment

To address the visual limitations identified in this benchmark, prospective agricultural inspection architectures should explore integrating RGB computer vision with complementary sensing modalities as promising research directions:

1. Hyperspectral and Near-Infrared Spectroscopy: Prospective integration of spectroscopy in the near-infrared and short-wave infrared ranges could probe internal fruit quality parameters such as moisture content, sugar concentration, and cellular breakdown, offering non-destructive assessment capabilities that extend beyond exterior RGB surface photography.
2. Volatile Organic Compound (VOC) and Chemical Sensing: As a prospective investigation area, specialized gas-sensor arrays and chemical sensing architectures can monitor ambient headspace for volatile markers associated with ripening, decay, or chemical adulteration, providing an orthogonal signal to optical classification [20].
3. Dynamic Cold-Chain Telemetry Integration: Future routing decision frameworks could investigate integrating image-based surface assessments with historical transit telemetry (temperature, relative humidity, and transit duration) to support predictive quality management throughout the cold chain. We emphasize that these multi-modal avenues represent prospective research directions rather than proven capabilities of the current offline vision benchmark.

7. Limitations and Threats to Validity

To maintain rigorous scientific transparency, this section explicitly outlines the methodological boundaries, data provenance constraints, and experimental threats to validity governing the reported findings.

7.1. Image Records versus Biological Specimen Independence

All empirical evaluations in this study were conducted strictly on digital image files ($N = 13,465$) retrieved from public open-access repositories. As documented in Section 3.2, public photographic archives do not provide specimen-level tracking identifiers. Consequently, multiple photographic records may represent burst captures, differing viewpoints, or sequential rotations of the same physical fruit. While perceptual hash auditing at radial thresholds $r \leq 8$ identified and excluded near-duplicate files across evaluation tiers (Section 3.6), mathematical hash pruning cannot prove complete biological specimen independence. All reported bootstrap confidence intervals and statistical summaries are strictly conditional on image-level resampling exchangeability.

7.2. Dataset-Reported Treatment Labels without Independent Chemical Assays

The condition annotations for Cohort B (treatment-labelled known shift) and Cohort C (treatment-labelled unseen shift) rely entirely on metadata published by the FruitVision repository creators [12], who

report procuring fruit subjected to formalin preservative immersion. No independent analytical chemistry, gas chromatography–mass spectrometry (GC-MS), high-performance liquid chromatography (HPLC), or destructive tissue assays were performed by the present authors to verify formaldehyde concentration, immersion duration, penetration depth, or residual tissue toxicity. Our findings establish strictly that on this specific dataset-reported cohort, the evaluated post-hoc vision scoring algorithms exhibited weak discrimination between images labelled as treated and natural known-class images; we do not claim to provide a verified chemical assay or toxicological validation.

7.3. Residual Within-Repository Confounding under Source Matching

Our source-stratified evaluation design (Section 3.5) reduces repository-level provenance differences, while within-repository acquisition variation remains possible. In crowdsourced or field-collected datasets such as FruitVision and Sultana et al., distinct fruit classes may have been photographed across different market stalls, on different calendar dates, under varying solar illumination, or by different field personnel using different handheld mobile cameras. Consequently, subtle background, lighting, or sensor differences between species subsets within a shared repository cannot be entirely ruled out.

7.4. External Acquisition and Environmental Variability

The external evaluation partitions reflect uncalibrated consumer-grade photography. The images lack radiometric calibration targets, standardized reflectance standards, uniform lux measurements, or controlled illumination geometry. While this unconstrained variability provides a realistic stress test of real-world intake conditions across diverse photographic environments, it limits the ability to isolate specific optical phenomena (e.g., separating surface specular gloss from subsurface light scattering).

7.5. Statistical Conditioning and Shared Baseline Non-Independence

All reported 95% confidence intervals were generated using non-parametric percentile bootstrap resampling ($B = 1,000$, seed 42) stratified across classes and condition groups. Although Cohorts A–C share the same underlying FruitVision known-class reference records ($N_{\text{id}} = 4,157$), bootstrap resampling was performed independently for each cohort evaluation. Therefore, the marginal confidence intervals should not be interpreted as paired intervals for differences between cohorts. These cohorts are not independent experimental replications, and their point estimates and confidence intervals share an underlying empirical reference sample.

7.6. Lack of Physical Sorting and Conveyance Validation

This study represents an offline computational benchmark on static digital imagery. The experimental protocol does not evaluate dynamic high-speed conveyor transport, motion blur, variable camera trigger latency, vibrating fruit orientations, or mechanical pneumatic ejection physics. Translating the Dual-Gate decision logic into operational industrial sorting lines will require embedded hardware benchmarking (e.g., NVIDIA Jetson or edge TPU acceleration) and physical pneumatic sorting validation.

7.7. Algorithmic Implementation Variants

As detailed in Section 4.3, two of the five evaluated post-hoc scoring algorithms represent specific implementation variants rather than exact reproductions of original literature:

1. ODIN-Style Perturbation Variant: Canonical ODIN computes perturbations using the cross-entropy gradient with respect to predicted class softmax probabilities. Our implementation evaluates an ODIN-style variant optimizing the sign gradient of the maximum temperature-scaled logit ($T = 1000.0, \varepsilon = 0.0014$), matching the verified execution script.
2. Modified OpenMax Heuristic: Our evaluated OpenMax implementation models Weibull tails on 1280-dimensional global pooled features and employs an empirical rank-weighting schedule ($w_{c_r} = 1 - \frac{r-1}{\alpha}(1 - \omega_r)$) that leaves the top-1 predicted logit unaltered ($\alpha = 2, \eta = 20$).

We report results strictly for these verified implementations and do not claim that they represent the theoretical limits of alternative parameterizations.

7.8. Scale Dependence of the Continuous Margin Proxy

Continuous ROC analysis for the Dual-Gate cascade relies on an exploratory linear margin proxy: $S_{\text{proxy}}(x) = (S_{\text{Energy}}(x) - \tau_E) + (S_{\text{MSP}}(x) - \tau_M)$. Because raw Energy scores are mathematically unbounded on

this unweighted linear sum is scale-dependent and dominated by Energy variance. The term $\rho > 0.992$ refers to the empirical Spearman rank correlation between the proxy and raw Energy scores across all 13,465 evaluated image records. All offline operational screening metrics (UFAR, UFAR-Fresh, KRR, Retention) are evaluated strictly using the discrete Boolean conjunction ($g_E(x) \wedge g_M(x)$) and are immune to proxy scale dependence.

8. Conclusion

This study conducted a rigorous, source-stratified empirical benchmark investigating the robustness of closed-set fruit-quality classification and post-hoc out-of-distribution detection under external domain, semantic species, and dataset-reported treatment-labelled condition shifts across $N = 13,465$ evaluated photographic records.

Our findings provide empirical answers to the three core research questions:

1. Resolution of RQ1 (External Domain Robustness): Evaluating a high-performing frozen MobileNetV2 classifier (96.92% [94.71%, 98.68%] top-1 accuracy on laboratory benchmarks) against external repository imagery reveals a substantial 28.56 percentage-point degradation to 68.36% [67.20%, 69.40%] top-1 accuracy (69.14% on FruitVision B, 65.63% on Sultana B). This degradation is marked by an asymmetric collapse on Fresh Apples (32.12% accuracy; 310/965 correct), with frequent misclassifications into decay classes (393 predicted Rotten Apple, 255 Rotten Orange out of 965). Qualitative visual inspection indicates natural surface striations, lenticel spots, and specular reflections consistent with varietal russeting; we present this texture correlation strictly as an untested observational hypothesis rather than a causally verified visual mechanism.
2. Resolution of RQ2 (Comparative Out-of-Distribution Discrimination): Under source-stratified semantic shift, Energy-based scoring (AUROC = 81.57% [80.81%, 82.38%] on FruitVision, 67.52% [65.67%, 69.25%] on Sultana) and Ledoit–Wolf Mahalanobis distance (81.41% and 68.15%) achieve superior separation compared to baseline Maximum Softmax Probability (72.42% and 59.77%). However, under dataset-reported chemical preservative shift (Cohort B), evaluated post-hoc scoring algorithms exhibit weak or near-chance discrimination: Energy AUROC is 53.13% [52.16%, 54.09%], the Energy–MSP linear-margin proxy (exploratory) is 53.96% [52.96%, 54.91%], MSP is 58.77%, and Ledoit–Wolf Mahalanobis distance is 47.91%, with retrospective FPR@95 exceeding 91% across all methods. On this evaluated treatment cohort, tested RGB post-hoc scoring methods did not reliably separate images recorded under the reported treatment protocol from natural fruit.
3. Resolution of RQ3 (Operational Feasibility and Rejection Trade-offs): Validation-calibrated decision gating achieves partial unknown screening on semantic shifts (UFAR = 16.07% in A, 9.54% in C), but imposes a substantial in-distribution abstention rate that could present operational challenges: the system rejects 42.43% [41.10%, 43.76%] of known-class image records across external repository conditions (KRR = 41.98% on FruitVision B, 44.01% on Sultana B), retaining automated decisions on only 57.57% of known-class image records. Crucially, 42.83% [41.26%, 44.34%] of dataset-reported treatment-labelled image records (821/1,917) are accepted by the Dual-Gate and assigned one of the classifier's Fresh-class output labels.

We emphasize that this investigation is an empirical diagnostic evaluation, not a claim of algorithmic novelty or a certified production sorting deployment. By identifying the visual failure boundaries of convolutional neural networks in post-harvest grading, this work underscores the critical necessity of source-stratified benchmarking protocols. Prospective agricultural automation research should explore transitioning beyond standalone RGB computer vision toward multi-modal architectures integrating complementary modalities such as hyperspectral imaging, headspace volatile organic compound sensing, and dynamic cold-chain telemetry.

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